

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for economic control with recursive utility, endogenous preferences, and strategic interaction. Evaluation, feasible improvement, and verification are separate maps. A guarded critic and signed interval action cover connect neural proposals to nodal economic error accounts. We additionally derive a continuous-time performance bound for a dense nonlinear capital economy. Consumption curvature and synchronous state comparison control arbitrary adapted deviations; a schedule-centred neural actor inherits a computable regret bound over the original, unrestricted comparison class. Interval spectral verification and scalar quadrature avoid state-space covering. Generic actors and direct HJB comparators are trained in dimensions ten, twenty, and fifty, with all seeds, deployment costs, and paired payoffs reported. The feasible schedule remains an explicit, cheaper comparator with a tighter structural bound. Preference refinement, financed welfare comparisons, and finite-game deviations retain their separate scopes. The results distinguish guaranteed policy performance from optimizer convergence and measured actor gains.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Quantitative economic models increasingly place several margins of adjustment inside the same dynamic decision problem. An investor chooses consumption and risk exposure while managing a preference state. A sophisticated consumer chooses today knowing that tomorrow's self will reconsider. A firm invests while anticipating the investment policies of its rivals. These problems share a computational structure: a continuation value must correctly evaluate a policy, and that policy must

satisfy the appropriate optimality or equilibrium condition. The two requirements are distinct. A perfectly evaluated policy need not be optimal, and an attractive policy evaluated by an inaccurate continuation value need not improve welfare.

This paper develops Neural Bellman Operators (NBO) around this distinction. A critic represents continuation utility. A feasible actor represents the controls that enter the economic generator. The critic is fitted to a policy-evaluation equation, including its terminal and boundary data. Holding the critic and its derivative channels fixed, the actor improves its own Hamiltonian. An independent maximization step measures the remaining feasible-action gap. The resulting error decomposition concerns economic values and deviations, rather than only parameter gradients or the scalar quantity minimized by an optimizer.

The motivation remains continuous-time economic control, including recursive utility and costly preference formation in the motivating model of [Qi \(2025\)](#). We do not replace these problems by a prediction task. Instead, we make explicit how a neural approximation is connected to the original decision problem. In the smooth case, the connection follows from a verification bound. In the nonsmooth case, it follows from a monotone Bellman approximation and a comparison principle. The actor and critic may be trained by different numerical procedures in either case. Their economic interpretation is determined by the equations they satisfy and by the errors that are verified after training.

The paper makes three connected contributions. First, it specifies a neural policy-evaluation and improvement map with explicit derivative routing, boundary conditions, and an indexed evaluation guard. The guard bounds a selected-policy target residual by construction; it does not claim convergence of Adam or an uncorrected network. Second, it makes the complete-action error account operational through signed interval derivatives and verified face domination, with natural union bounds at stopping switches. A matched enclosure experiment isolates this calculation from policy training. Third, it carries the accounts through the original preference, capital, and strategic applications: nested-grid retraining, a model-specific compensating transfer, an absolute continuous-time capital benchmark, and finite dynamic best responses. Comparison, monotonicity, interval inclusion, and approximate policy iteration remain inherited principles. The new results concern the specified guarded

map, its verified output, and its measured economic accuracy, not an unqualified claim that a neural actor dominates classical or actor-free alternatives.

The numerical program uses generic multilayer critics and actors as well as clearly separated analytical laboratories. The finite preference and game studies train actors using exhaustively constructed continuation-payoff labels; their results establish neural policy representation, not avoided maximization. The continuous-action preference study instead differentiates the actor’s own payoff, records local improvement as a hybrid policy, and reserves global action covering for independent verification after training. Direct optimization and an actor-free search ablation expose which parts of the computation produce the measured policy quality. Additional gated and projection comparisons exploit the same economic problem rather than weakening the baseline. Dense nonlinear capital experiments use paired pathwise payoffs to diagnose state coverage. Homothetic portfolios, temporal-self formulas, the nonsmooth exit problem, and the exact quadratic family remain distinct economic and algebraic checks. No matrix calculation is counted as neural training, and no grid refinement difference is silently converted into a continuous-time bound.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing

action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion (Epstein and Zin, 1989, Duffie and Epstein, 1992). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; Herdegen et al. (2021) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of Björk et al. (2016), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions (Borkar, 2024). They do not convert constant-step Adam into a convergence theorem for economic optimality.

The continuous-capital result closes a different part of the error account: curvature and a verified coupling norm bound the gain from arbitrary adapted deviations, while a schedule-centred actor supplies a uniformly constrained neural policy. The resulting bound concerns the original diffusion rather than a nodal surrogate. Its stronger structural benchmark is reported alongside every fitted policy.

Section 2 specifies the economy. Section 3 defines the implemented map and Section 4 establishes its error accounts. Section 5 completes the differential theorem–algorithm–experiment chain. The subsequent sections retain and extend the portfolio, recursive-utility, preference-adjustment, temporal-self, dynamic-game, and scaling applications. The appendix proves the principal results. The supplement gives arithmetic details, application-specific assumptions, every seed-level comparison, and sensitivity results.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a,t,x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-

state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this revision uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} |V(S_T)|] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section 8.

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The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_\xi(t, x) = G(x) + (T - t)N_\xi(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_\xi = G_B + d_B N_\xi$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_\nu \left[-\partial_t v_{\bar{\xi}} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\bar{\xi}}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_\nu \mathcal{H}^{a_\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective used in the reviewed draft satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (8)$$

Thus the revised critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a_\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv} D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (9)$$

Algorithm 1 NBO evaluation, improvement, and acceptance

Require: Model, admissible actions, utility domain, boundary data, training and validation domains.

1: Initialize a feasible actor and a boundary-compatible critic.

2: **for** each outer iteration **do**

3: Freeze actor outputs. Update only the critic using (6) or a positive-weight Bellman evaluation residual.

4: Freeze the critic and every derivative channel. Improve only the actor using (7) or a feasible action solver.

5: Evaluate boundary error, policy-evaluation error, and the independent feasible-action gap.

6: Record the initialization, updates, conditioning, unsuccessful attempts, and stopping rule.

7: **if** the declared economic error criterion is met **then**

8: Return the policy, value, verification domain, and error report.

9: **end if**

10: **end for**

11: Return the computed approximation and its measured errors; do not infer optimality from optimizer stationarity.

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

3.3. Conditioning and stopping

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that

entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do

not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections 5 and 8.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action’s generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in A_i} (T_n^a w)_i. \quad (11)$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION 1—Budget-safe action and policy envelopes: *Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,*

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (12)$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the

interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate's precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpolation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (13)$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 2—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,*

$$|v(t, x) - V^a(t, x)| \leq e^{Ls}B + A_L(s)e, \quad (14)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls}B + A_L(s)(e + \eta), \quad (15)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls}B + A_L(s)(2e + \eta). \quad (16)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled residual is not yet the uniform hypothesis of the theorem. Section 4.3 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (17)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$ gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 3—Discrete NBO certificate: *Suppose (17) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (18)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 4—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 3. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy*

$$\delta_h + \epsilon_h = o(h), \quad (19)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural.

The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

4.2. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (20)$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$ exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

4.3. Coverage and independent maximization

A verification distribution is declared independently of the actor's occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} , then

$$\|r\|_{\infty} \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (21)$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

4.4. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + hr_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION 5—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (22)$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 4.

5. A TRAINED NEURAL BOUNDARY-VALUE PROBLEM

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (23)$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_\xi(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_\xi\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (24)$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION 6—Continuous-domain economic certificate: *For a stopped discounted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissible. Suppose throughout D that*

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_\infty \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_\infty \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (25)$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v, v', v'' , and the actor on every leaf. Equation (24) supplies a global action enclosure. We also use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (26)$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

5.1. Controlled comparison and failures

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative

TABLE I
MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[.02, .6]$. All six certificates have uniform evaluation error at most .002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly. All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation.

Fitting and full-domain verification are separately timed because their computational burdens are different.

6. ANALYTICAL AND NONSMOOTH BENCHMARKS

The time-additive Merton model has wealth dynamics

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma X dW, \quad U(c) = \frac{c^{1-\gamma}}{1-\gamma}.$$

For the stationary infinite-horizon benchmark, set $c = mX$, $V = KX^{1-\gamma}/(1-\gamma)$, and impose the usual admissibility and transversality requirements. Writing

$$p^* = \frac{\mu - r}{\gamma\sigma^2}, \quad R_* = r + p^*(\mu - r) - \frac{1}{2}\gamma(p^*)^2\sigma^2,$$

substitution and maximization give

$$m^* = \frac{\rho + (\gamma - 1)R_*}{\gamma}, \quad K = (m^*)^{-\gamma}. \quad (27)$$

With $\rho = .04$, $\gamma = 2$, $\mu = .08$, $r = .02$, and $\sigma = .2$, these numbers are $p^* = .75$ and $m^* = .04125$. These are stationary constants; an arbitrary finite-horizon problem with an unspecified bequest does not have the same constant consumption rule. A finite-horizon verification test can use the matching bequest $KX^{1-\gamma}/(1-\gamma)$, explicitly declared rather than silently omitted.

For $\mu = .01$ with no short sales, the unconstrained share is $-.125$ and the constrained optimum is zero at every positive wealth. Its effective return is r and $m^* = .03$. The maximized HJB remains an equality. The action KKT condition is an inequality at the boundary, but it does not turn this regular-control problem into an obstacle problem or create a wealth-dependent kink. The exit experiment in Section 4, rather than this homothetic corner solution, supplies the nonsmooth selection test.

6.1. Executed calibration and interpretation

The replication program trains a homothetic neural subfamily: a positive value coefficient, a sigmoid consumption-share output, and a projected portfolio-share

TABLE II
CORRECTED PORTFOLIO CALIBRATION: ANALYTICAL TARGETS AND EXECUTED BLOCK UPDATES

Model	Target m	Target p	Runs	Maximum policy error
Time-additive Merton	.04125	.75000	3	5.6×10^{-16}
No-short-sale corner	.03000	.00000	3	2.3×10^{-16}
Recursive matching-bequest test	.04550	.30000	3	7.0×10^{-17}

Policy error is the maximum absolute error in the two constant policy outputs, maximized over all three seeds. Calculations use CPU float64 in the declared homothetic subfamily. Machine-level calibration accuracy is not evidence about an unrestricted network, a GPU runtime, or a general recursive infinite-horizon problem.

output. These are neural output parameters in a known invariant representation, not an unrestricted multilayer approximation. The two Adam optimizers have learning rates .025 and .008; each outer iteration contains three critic updates and one actor update. There are 1,600 outer iterations and seeds 0,1,2. All attempted runs, including their coefficient and policy traces, are saved. These calibrations optimize the stationary coefficient equation and then check the resulting coefficient against the matching bequest; the coefficient error is a boundary error in that finite-horizon interpretation, not a boundary condition hard-imposed during these particular updates.

Table II reports quantities generated by the new code. The normalized policy-evaluation residual is below 4×10^{-16} in these runs. These results establish that the revised graph and formulas agree in the exactly representable laboratory. They do not establish a broad advantage over a direct analytical calculation. In the exit laboratory, grids with 40, 80, 160, and 320 intervals recover the grid value to within 3×10^{-15} ; the wrong smooth sequence has scaled residual between 1.984 and 1.999. All grid nodes, not a selected interior sample, enter that diagnostic.

7. RECURSIVE UTILITY

Let $a = 1 - 1/\psi$, $b = 1 - \gamma$, and $Y = bV > 0$, with $\psi \neq 1$ and $\gamma \neq 1$. We use the difference-form aggregator

$$f(c, V) = \frac{\rho}{a} \left[c^a (bV)^{1-a/b} - bV \right]. \quad (28)$$

Its consumption derivative is

$$f_c = \rho c^{a-1} (bV)^{1-a/b} > 0. \quad (29)$$

When $\gamma = 1/\psi$, one has $a = b$ and $f = \rho(c^b/b - V)$, a positively normalized time-additive CRRA aggregator. The factor b must not multiply the consumption term outside its power. Doing so reverses consumption monotonicity when $\gamma > 1$. This distinction follows the difference-form convention discussed by [Herdegen et al. \(2021\)](#).

For a homothetic candidate $V = KX^b/b$, the maximization equation becomes

$$0 = \sup_{m,p} \left\{ \frac{\rho}{a} [m^a K^{-a/b} - 1] + r + p(\mu - r) - m - \frac{1}{2} \gamma p^2 \sigma^2 \right\}. \quad (30)$$

The consumption first-order condition is $\rho m^{a-1} K^{-a/b} = 1$. Substitution yields

$$p^* = \frac{\mu - r}{\gamma \sigma^2}, \quad m^* = \psi \rho + (1 - \psi) \left[r + \frac{(\mu - r)^2}{2\gamma \sigma^2} \right], \quad K = [\rho(m^*)^{a-1}]^{b/a}. \quad (31)$$

For the retained parameters $\gamma = 5$, $\psi = 1.5$, $\rho = .04$, $\mu = .08$, $r = .02$, $\sigma = .2$, this gives $(m^*, p^*) = (.0455, .3)$.

7.1. A well-defined validation problem at the retained parameters

The numerical calibration uses (31) as a finite-horizon matching-bequest test, not as an unqualified infinite-horizon theorem. Choose $T > 0$, stop wealth on leaving $[x, \bar{x}]$ with $0 < x < \bar{x} < \infty$, and impose the candidate Kx^b/b on the stopping boundary and at T . Consumption shares are in a compact positive interval containing .0455, and portfolio shares are in a compact interval containing .3.

For these retained parameters, $a > 0$ and $b < 0$. Since c ranges over a compact positive interval, there exist $0 < \underline{Y} < \bar{Y} < \infty$ such that $f(c, \underline{Y}/b) \leq 0$ and $f(c, \bar{Y}/b) \geq 0$ for every feasible c . Indeed, $c^a Y^{-a/b}$ tends to zero as $Y \downarrow 0$ and to infinity as $Y \uparrow \infty$. Enlarge the interval so that it contains the boundary candidate. The constants $\bar{Y}/b$ and $\underline{Y}/b$ are respectively lower and upper barriers for the backward utility equation. Scalar BSDE comparison therefore keeps every policy utility in this interval. The aggregator is Lipschitz there. The smooth homothetic candidate maximizes the

Hamiltonian and matches all stopping payoffs; Theorem 2 identifies it as the value of this specified finite-horizon problem.

The choice of a matching bequest is an analytical validation device. It does not establish that the same candidate is selected by every infinite-horizon recursive formulation. A separate infinite-horizon application must check its utility-process class and transversality, particularly when risk aversion and intertemporal-complementarity parameters straddle one (Herdegen et al., 2021). The paper retains that economic application, but does not use a scalar root to bypass its selection issue.

A critic can enforce $bV > 0$ through $bv_{\xi} = \exp(N_{\xi})$. With a positive terminal target, logarithmic interpolation between bG and the neural output enforces both the terminal condition and the power domain. Additional barriers or bounded transformations enforce a declared invariant interval. Such transformations are stated in the architecture, not imposed retrospectively after a failed evaluation.

7.2. Stability and baseline design

The recursive experiment in Table II tests the corrected sign, coefficient, and block graph. It does not compare independent DDPG training trajectories. The reviewed manuscript’s constructed oscillating curve and manually entered coordinates remain in the archival source, but have no evidentiary role in this revision. A matched neural comparison must use the same utility process, domain, bequest, action restrictions, training budget, and held-out evaluation quantities. It must report all failed seeds and value-domain violations. Neither a zero-centered target nor three successful homothetic calibrations establishes general relative stability.

8. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

The endogenous-preference application retains its internal adjustment margin. Let u be risk aversion, X wealth, and θ adjustment effort. On a declared interval $[\underline{u}, \bar{u}] \subset (1, \infty)$, specify the reflected process

$$du = \theta dt + \sigma_u dW^u + dL^{\underline{u}} - dL^{\bar{u}}, \quad (32)$$

$$dX = [(r + p(\mu - r))X - c]dt + p\sigma_X X dW^X, \quad d\langle W^u, W^X \rangle_t = \varrho dt. \quad (33)$$

The nondecreasing local-time regulators act only at their respective boundaries. This supplies the state mechanism that a bounded adjustment actor does not supply. The numerical reference uses $[1.2, 3]$, so the utility singularity at $u = 1$ is never reached. In an alternative positive-state model one could specify a multiplicative or transformed diffusion, but that would be a different primitive and must be identified as such.

8.1. Cardinal utility and the objective

Fix a reference consumption unit $\bar{c} > 0$. The baseline flow utility is

$$U(c, u) = \frac{(c/\bar{c})^{1-u}}{1-u}, \quad r_0(c, u, \theta) = U(c, u) - \frac{1}{2}k\theta^2. \quad (34)$$

The reference calculation takes $\bar{c} = 1$, preserving the unnormalized CRRA family of the original application. The agent maximizes

$$\mathbb{E}_{t,u,X} \left[\int_t^\tau e^{-\rho(s-t)} r_0(c_s, u_s, \theta_s) ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (35)$$

Both the payoff and the adjustment cost have the same time-relative discount convention. The terminal/exit payoff and the state domain are stated for each calculation.

A useful family for normalization sensitivity is

$$U_\kappa(c, u) = \frac{(c/\bar{c})^{1-u} - 1}{1-u} + \kappa(u).$$

The baseline corresponds to $\kappa(u) = 1/(1-u)$. The log-continuous normalization corresponds to $\kappa = 0$ and is an economically different preference-formation model. A calibration of κ fixes the cardinal value of a reference consumption stream at each preference state. The choice is not identified by risk-aversion curvature alone.

In particular,

$$U_u(c, u) = \frac{(c/\bar{c})^{1-u} \{1 - (1-u) \log(c/\bar{c})\}}{(1-u)^2}. \quad (36)$$

For $c/\bar{c} > 1$ and $u > 1$ this derivative is positive. At $c = 2$, $u = 2$, $\bar{c} = 1$, it is approximately .84657. This is a derivative of flow utility, not a sign theorem for the

endogenous value derivative V_u . The agent's full incentives also depend on future consumption, boundary behavior, and the adjustment technology.

8.2. Hamiltonian, constrained policies, and economic implications

The HJB equation is

$$\begin{aligned} -V_t = \sup_{c,p,\theta} \{ & U(c,u) - \frac{1}{2}k\theta^2 - \rho V + \theta V_u + [(r + p(\mu - r))X - c]V_X \\ & + \frac{1}{2}\sigma_u^2 V_{uu} + \frac{1}{2}p^2\sigma_X^2 X^2 V_{XX} + \varrho\sigma_u\sigma_X p X V_{uX} \}. \end{aligned} \quad (37)$$

Reflection gives $V_u = 0$ at the two preference boundaries, in the appropriate boundary-value sense. The correlation term is retained with its correct coefficient.

For an interior optimum with $V_X > 0$ and $V_{XX} < 0$, the first-order conditions are

$$c^* = \bar{c} [\bar{c} V_X]^{-1/u}, \quad (38)$$

$$\theta^* = V_u / k, \quad (39)$$

$$p^* = -\frac{V_X(\mu - r)}{X V_{XX} \sigma_X^2} - \frac{\varrho \sigma_u V_{uX}}{X V_{XX} \sigma_X}. \quad (40)$$

For independent box constraints, strict concavity in the corresponding action permits projection of these unconstrained maximizers onto their intervals. If V_{XX} has the wrong sign or constraints couple the actions, a joint feasible maximization is required instead. The formula for p must not be used where its denominator vanishes.

Equation (39) gives a conditional shadow-price interpretation of internal adjustment. Holding V_u fixed, a larger k reduces its magnitude. Across economies, however, V_u itself depends on k , so this is not a global policy comparative static. A valid unconditional implication is that the optimized value is weakly decreasing in k : every fixed admissible policy loses $\frac{1}{2}(k_2 - k_1)\mathbb{E} \int e^{-\rho s} \theta_s^2 ds$ when the cost rises. When an envelope derivative exists,

$$\partial_k V^k = -\frac{1}{2}\mathbb{E}^{a^k} \int_t^\tau e^{-\rho(s-t)} \theta_s^2 ds \leq 0. \quad (41)$$

The sign of the hedging component in (40) remains conditional on V_{uX} and the curvature of wealth value. A negative correlation alone does not determine it.

8.3. Independent two-state reference and identification

The executable reference solves the finite-horizon problem with $T = 1$, $u \in [1.2, 3]$, wealth stopped at $[.5, 2]$, and $G(u, X) = X^{1-u}/(1-u)$ at exit and terminal time. Parameters are $(r, \mu, \sigma_X, \sigma_u, \varrho, \rho) = (.02, .08, .2, .08, -.3, .04)$. The finite action set is the product of $c/X \in \{.02, .05, .1\}$, $p \in \{0, .375, .75\}$, and $\theta \in \{-.15, 0, .15\}$. A four-point Brownian cubature with positive interpolation weights performs backward induction. The wealth boundary is imposed at every time step; preference overshoots are reflected. This reference defines a finite-action numerical approximation, not continuous-action optimality.

At $(u, X) = (2.1, 1)$, the original 17×25 state grid and 20 time steps give values -11.915211 and -11.956983 for $k = 1$ and $k = 5$. The 25×37 grid and 40 time steps give -11.866768 and -11.908738 . Both select $(c/X, p, \theta) = (.1, .75, -.15)$ at that point. The approximately .048 refinement change is reported separately from training error. The higher-cost value is weakly lower at every recorded node and time. These original references remain useful consistency checks; Section 8.4 supplies actual neural fits and shows how strongly the finite action domain affects their economic comparison.

The model also specifies what a structural estimation exercise would require. For observations $Y_j = \mathcal{M}(S_{t_j}, a_{t_j}; \vartheta) + \varepsilon_j$, let $m(\vartheta)$ denote selected population moments of the induced observation process. Local identification requires a rank condition on $D_\vartheta m$, after accounting for latent preference states, normalization parameters, and observation noise. Variation of a simulated policy with k alone is insufficient. Numerical value and policy errors must additionally be propagated into moment errors. The endogenous-preference application is therefore retained as an economic model with an explicit internal margin, not used as evidence that its structural parameters have already been identified from data.

The next comparison is the retained R8 evidence under its original enclosure. The new signed envelope and fresh nested-grid policies are reported separately in Section 13; historical failure counts are not overwritten.

8.4. Neural preference policies and the role of the action oracle

We compute the original preference-adjustment economy with the cardinal utility, shock correlation, reflected preference state, and wealth stopping payoff specified above. The main grid has 17 preference points, 25 log-wealth points, and 20 time steps over a unit horizon. Adjustment cost is $k = 1$. The expanded control box is

$$m = c/X \in [.02, .30], \quad p \in [0, 1.5], \quad \theta \in [-.45, .45].$$

The inherited finite-action comparison uses the 175 triples formed by five consumption shares, five portfolio shares, and seven adjustment efforts, including all endpoints. This is the same constrained economy across the matched finite methods, not a change of utility normalization.

Two experiments distinguish action representation from action optimization. First, the R7 finite study trains two-hidden-layer, width-32 tanh critics and categorical actors. At each backward step it computes the complete 175-action payoff vector using the current continuation critic. Its exact argmax provides a label, and the vector also supplies the actor’s expected action-loss objective. Thus *the training oracle is exhaustive*, even when the final actor is accurate. It is a useful neural policy-representation experiment, but not evidence that the actor avoided Bellman maximization.

Second, R8 trains a continuous-output actor directly through its own positive-weight payoff. It never receives an all-action value vector or an exact argmax label. The actor receives 160 Adam updates per time level; the continuation critic receives 320 Adam updates and up to 300 L-BFGS iterations, with failed polishing steps rolled back. The actor-free direct comparator optimizes eight independent bounded controls at every state for the same 160 iterations, followed by the same critic fit. Every restart and rejected action query is charged. Neither procedure invokes a Ric-

cati solution, an analytically invariant policy class, or the independently computed reference during training.

Finite-action neural policies. The raw R7 actor is the primary neural representation result. Its full dynamic policy loss is below the declared .1 target for each fixed seed. A top-four shortlist and a separately reported correction map further reduce the loss. The correction map is not part of the raw network. The complete finite policy, not only one-step deviations against a fitted critic, is evaluated by independent backward recursion. These results repair the earlier R6 actor failures, which remain

TABLE III

FINITE PREFERENCE POLICIES: R7 EVIDENCE

Method	Seed	Raw loss	Final loss	Value error	Seconds	Guard (%)
NBO	11	0.0445	0.0261	0.0924	30.59	0.294
Direct	11	0.0181	0.0181	0.0790	24.79	0.000
NBO	29	0.0500	0.0260	0.0814	30.49	0.332
Direct	29	0.0263	0.0263	0.0551	25.19	0.000
NBO	47	0.0500	0.0263	0.1325	29.89	0.230
Direct	47	0.0264	0.0264	0.0824	25.17	0.000

Loss bounds cover every stored state and time of the same 175-action economy. Seconds include training and the three reported policy verifications; classical backward induction averages 0.102 seconds. A raw actor, shortlist, and corrected policy are distinct objects. Utility losses are absolute units.

in the historical numerical record. They do not reverse the cost comparison: the matched direct finite-Bellman fit is faster and more accurate on average, and classical backward induction is substantially cheaper on this small grid. Tables report those comparisons without selecting a favorable seed. The actor’s approximation cost is part of the result, not an argument for discarding the economic application.

Continuous-action optimization without exact labels. The initial continuous actor and direct-gradient pilot fail the .1 deviation diagnostic. The actor can saturate at a portfolio bound while a lower portfolio share improves continuation value. Four actor heads with output reinitialization do not cure that failure in the retained

pilot. We therefore execute a disclosed local-improvement variant: compare the actor proposal with eight box corners, then perform two coordinate-search sweeps at each of eight decreasing step sizes. The same local search is applied to the direct comparator. An actor-free ablation omits all gradient action updates and starts the identical search from feasible proposals. This ablation measures how much of the hybrid result comes from the search itself. The table distinguishes

TABLE IV
CONTINUOUS-CONTROL TRAINING WITHOUT EXACT MAXIMIZING LABELS

Method	Seed	Raw diagnostic	Hybrid gain	Train (s)	Queries (10^6)
Actor + search	11	0.2596	0.0617	15.01	2.261
Actor + search	29	0.0733	0.0188	28.49	2.261
Actor + search	47	0.2597	0.0475	26.46	2.261
Direct + search	11	0.1554	0.0303	32.02	11.841
Direct + search	29	0.1566	0.0203	32.80	11.841
Direct + search	47	0.1554	0.0251	30.52	11.841
Search only	11	5.5370	0.0617	13.97	0.961
Search only	29	7.2093	0.0616	13.72	0.961
Search only	47	8.0316	0.0616	12.87	0.961

Raw diagnostic is the largest finite-reference payoff minus the raw policy payoff. Hybrid gain is the augmented finite-deviation gain, a lower bound on continuous-action regret, not a certificate. All methods use zero exhaustive training backups and zero exact argmax training labels. Search-only omits action-gradient updates. Training, reference, and full-cover verification costs are separate.

three quantities. The raw-network loss against the finite reference tests the actor alone. The augmented finite-deviation gain permits all 175 inherited actions and the learned continuous action at every state and time; it is a *lower bound* on missed continuous-action gains. The complete action-box upper bound in Proposition 1 is a separate verification result. A small augmented gain is not inserted into that proposition as if it were a uniform upper bound. Local changes occur at nearly every active state-time node in the principal hybrid runs, so this experiment does not describe a sparse correction map or an actor-only controller. The action cover includes binding constraints and wealth exits. Its unresolved boxes remain in the

TABLE V

FULL CONTINUOUS-ACTION POLICY BOUNDS ON THE TWO-STATE NODAL ECONOMY

Policy	Seed	Uniform bound	Center bound	Eval. (s)	Bound $\leq .1$
Actor + search	11	0.1294	0.0601	0.22	No
Actor + search	29	0.1227	0.0583	0.22	No
Actor + search	47	0.1301	0.0608	0.25	No
Direct + search	11	0.1153	0.0604	0.23	No
Direct + search	29	0.1156	0.0586	0.22	No
Direct + search	47	0.1203	0.0609	0.25	No
Search only	11	0.1196	0.0647	0.24	No
Search only	29	0.1179	0.0586	0.23	No
Search only	47	0.1209	0.0610	0.22	No

The full action box is covered at all 425 states and 20 times. The common optimal upper envelope costs 1624.59 seconds across the retained implementations and 40.96 million action-box evaluations. Each row adds its own interval policy evaluation. Unresolved leaves retain their upper bounds. No continuous-state/time error is set to zero.

bound when the declared budget is exhausted. The resulting upper array can audit the actor, the direct optimizer, and the local-search ablation without retraining. This produces an operational error account on a multidimensional state grid with a continuous, nonconcave action objective. It does not establish that all runs meet the tighter .1 complete-action target, nor that the actor dominates the direct method at matched economic accuracy.

Alternative representations and economic units. Additional same-model comparisons use a gated neural continuation architecture and tensor Chebyshev projection. The gated method uses DGM-style state-dependent gates (Sirignano and Spiliopoulos, 2018), but here it is fitted to finite Bellman targets, not called a mesh-free differential DGM solver. The projection exploits the same two-state geometry and imposes the same wealth-boundary lifting. Degrees, parameter counts, all-action costs, failures, and actual accuracy appear in the supplement. These are executed post-review comparisons, not results attributed to the earlier unexecuted development protocol. A generic semilinear BSDE comparator is not substituted for this

controlled-covariance, stopped problem without reformulating its second-order control term.

All losses are in the specified cardinal utility units. The optimal finite-model value at the center is approximately -4.8410 , and the initial value range is approximately $[-9.1599, -.1250]$. A loss of .1 is about 2.066 percent of the center's *absolute utility magnitude*; this scaling is not a consumption-equivalent welfare percentage. Endogenous curvature and adjustment costs make a common consumption-equivalent conversion a separate economic calculation.

Action bounds and controlled refinements. The inherited sensitivity study retains both adjustment-cost values, the original 27-action truncation, 175 and 1,053 actions in the expanded box, a wider action box, wider wealth bounds, and finer space-time grids. Consumption and portfolio upper bounds remain frequently active. Expanding the box changes the constrained economy; refining actions inside one box does not. R8 additionally evaluates the *same frozen continuous policy*, bilinearly interpolated in state and held on its original time schedule, on three nested state-time levels. This isolates changes in policy evaluation from retraining. The supplemental differences are observed refinement diagnostics, not an estimated order promoted to a continuous-time error bound. The original continuous control problem, its utility normalization, and its comparative-static questions remain the economic objects of interest.

9. SOPHISTICATED TEMPORAL SELVES

To define present bias in continuous time, fix the discount kernel

$$h(s) = \beta e^{-\rho s} + (1 - \beta)e^{-(\rho+\kappa)s}, \quad 0 < \beta \leq 1, \quad \kappa > 0. \quad (42)$$

It satisfies $h(0) = 1$ and discounts the immediate future more heavily than the distant future when $\beta < 1$. It is a finite-duration continuous-time specification; it is not the assertion that multiplying current utility by β creates present bias.

For a candidate Markov policy a^* , define the two continuation functions

$$W_j(t, x) = \mathbb{E}_{t,x}^{a^*} \left[\int_t^T e^{-\rho_j(s-t)} u(a_s^*) ds + e^{-\rho_j(T-t)} G(S_T) \right], \quad (\rho_1, \rho_2) = (\rho, \rho + \kappa). \quad (43)$$

Write $V = \beta W_1 + (1 - \beta) W_2$. Each component evaluates the *same* future policy:

$$0 = (W_j)_t + \mathcal{L}^{a^*} W_j + u(a^*) - \rho_j W_j, \quad W_j(T) = G. \quad (44)$$

The current self's spike-deviation condition is

$$a^*(t, x) \in \arg \max_{a \in \mathcal{A}(t, x)} \{u(a) + \mathcal{L}^a V(t, x)\}. \quad (45)$$

The deviations change both current flow utility and the transition of the continuation state. They do not hold all dynamic effects at the equilibrium action. The appendix derives (45) from a deviation on $[t, t + \epsilon]$ followed by a^* , thereby fixing the equilibrium concept.

A temporal-self NBO uses two policy-evaluation critics for (44) and one actor for (45). The current actor does not optimize the sum of future selves' parameter objectives. A target actor can stabilize a numerical iteration, but its agreement with the current actor is not an equilibrium certificate. The relevant diagnostics are component evaluation errors and one-shot deviation gains. The benchmark $\beta = 1$ collapses to the usual exponentially discounted problem.

9.1. An equilibrium calculation with endogenous continuation values

For deterministic wealth $dX = (rX - c)dt$, CRRA utility with $\gamma > 1$, and an infinite horizon, consider $c = mX$. Set $b = 1 - \gamma$ and $\lambda_j(m) = \rho_j - b(r - m) > 0$. Then

$$W_j(X) = \frac{m^b}{\lambda_j(m)} \frac{X^b}{b}, \quad 1 = m \left[\frac{\beta}{\lambda_1(m)} + \frac{1 - \beta}{\lambda_2(m)} \right]. \quad (46)$$

The expression on the right is strictly increasing in m on its admissible interval, starts at zero, and diverges at the first pole for $\beta > 0$. The equilibrium root is unique in this homothetic class. Its derivative with respect to β is negative because $1/\lambda_1 > 1/\lambda_2$.

Thus stronger present bias raises consumption in this specified equilibrium, not merely at a fixed shadow price.

With $r = .02$, $\rho = .04$, $\gamma = 2$, and $\kappa = .2$, the roots for $\beta = 1, .9, .7, .5$ are respectively .030000, .031372, .034606, and .038760. The code independently optimizes finite-duration deviations and records the gain divided by duration as the duration shrinks. For $\beta = .7$, this quantity declines from approximately 4.60×10^{-5} at duration .1 to 4.66×10^{-9} at duration .001. This is a diagnostic of the derived spike equilibrium. It does not assert that every finite-duration commitment deviation is unprofitable, which would be a different solution concept.

10. DYNAMIC GAMES

The dynamic Cournot application retains firm-specific capital and investment. Firm i has

$$dK^i = (I^i - \delta K^i)dt + \sigma_K K^i dW^i, \quad q_i = AK^i,$$

and payoff rate

$$r_i = P_N(K)q_i - I^i - \frac{1}{2}b_i(I^i)^2, \quad P_N(K) = \max \left\{ 0, P_0 - \frac{1}{M_N} \sum_j q_j \right\}.$$

For each experiment, the market-size normalization M_N , capital domain, investment constraints, covariance, terminal/liquidation values, and horizon are part of the primitives. Keeping $M_N = 1$ is a fixed-market comparison. Taking $M_N = N$ is a growing-market comparison. They do not have the same competitive-limit interpretation.

Given rivals' Markov policies a^{-i} , a player's best-response equation maximizes only its own action. In the stationary additive case,

$$\rho_i V_i = \sup_{a_i \in A_i} \{r_i(x, a_i, a^{-i}(x)) + \mathcal{L}^{a_i, a^{-i}} V_i\}. \quad (47)$$

A Markov-perfect equilibrium is a feasible profile satisfying these simultaneous unilateral conditions, with the same boundary and admissibility conventions. A zero policy-evaluation residual can hold for any consistently valued profile and therefore does not identify this equilibrium.

10.1. *Player-specific differentiation and deviation bounds*

Player i 's critic is trained with every policy frozen. Player i 's actor then differentiates only

$$-\mathbb{E}_\nu \mathcal{H}_i^{a_i(\phi_i), \text{sg}(a^{-i})} [\text{sg}(V_i), \text{sg}(DV_i), \text{sg}(D^2V_i)]. \quad (48)$$

The updated parameter is ϕ_i , not the full vector of all players' parameters. Rival actions are detached, and player j 's loss never contributes to player i 's actor gradient. An explicit regression test exercises this graph.

For each player, define the unilateral Hamiltonian gap by maximizing $\mathcal{H}_i$ over a_i while keeping rivals fixed. Applying Theorem 2 to that best-response problem bounds its finite-horizon exploitability by $2e^{L_i T} B_i + A_{L_i}(T)(2e_i + \eta_i)$. In a discounted finite-state game, Theorem 3 gives $(2\delta_i + \epsilon_i)/(1 - q_i)$. These are player-specific bounds; a sum of critic losses is not a substitute. They remain valid without uniqueness or convergence of the simultaneous policy-update dynamics, provided the reported profile satisfies the stated errors and best-response comparison hypotheses.

The static Cournot diagnostic makes the distinction visible. With payoffs $q_i(1 - q_i - q_j)$, symmetric Nash output is $1/3$ per firm. Joint-profit maximization gives a symmetric allocation $1/4$. At the latter profile, a firm optimally deviates to $3/8$, earning a gain of $1/64$. A joint objective can therefore be perfectly optimized and still fail a unilateral equilibrium test. This arithmetic is included as a mandatory regression check.

10.2. *Symmetry and market scaling*

Identical initial weights do not prove equilibrium uniqueness or permutation equivariance. An equivariant actor can be specified using own capital and a permutation-invariant aggregation of rival states. Even then, one must check unilateral deviations rather than infer equilibrium from architectural symmetry. Multiple starting profiles and unsuccessful best-response searches belong in a numerical report.

Under fixed market size, positive constant capital per firm cannot be called the competitive limit while total output tends to infinity and the truncated price tends to zero. Maintaining that capital entails investment and adjustment costs. Under $M_N = N$, positive capital per firm may instead be compatible with a growing market, but it describes a different sequence of economies. The original unsupported capital table remains in the archive. Section 10.3 replaces its evidentiary role with a fully specified finite dynamic game and independent best-response calculations, without asserting convergence of unrestricted neural game training.

10.3. *Neural dynamic capital policies and full unilateral deviations*

For two firms, we retain the capital game with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[\cdot 05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. Baseline adjustment coefficients are $b_1 = b_2 = 1$; the earlier asymmetric experiment has $b_2 = 1.2$. The principal neural grid has 17×17 states, 30 time steps, and 11 actions per player. Positive cubature and interpolation define the finite game. Market normalizations $M_2 = 1$ and $M_2 = 2$ are distinct economies, not evidence for an uncomputed large-population limit.

At each time and state, the R7 calculation trains separate critics and actors. It constructs both full 11×11 continuation-payoff matrices. The training labels are a pure one-step equilibrium when one exists and a minimum-maximum-defect pair otherwise. The actor's own objective is differentiated with the rival action and continuation critic fixed. Thus this study actually trains neural strategic policies, but it also pays for an exhaustive joint-action training oracle. Raw policies, three-by-three shortlist policies, and any corrected policies are distinct stored arrays.

After freezing the entire profile, an independent backward optimization permits every own action at every state and time against the fixed rival policy. Its value minus the profile payoff is the unilateral exploitability map. This is not a one-period test with the approximate continuation held fixed. The raw and final profiles meet the .1 absolute-profit target in every principal R7 case; the full-oracle guard is unused in those cases. Five state-time nodes across the six neural runs lack a pure one-step equilibrium for the fitted continuation matrices. Their locations

TABLE VI

R7 NEURAL DYNAMIC-GAME PROFILES AND FULL BEST RESPONSES

Market	Seed	Raw max	Firm 1	Firm 2	Missing	Seconds
1	11	0.0259	0.0164	0.0252	3	52.43
2	11	0.0241	0.0255	0.0163	1	56.13
1	29	0.0300	0.0288	0.0296	0	52.44
2	29	0.0626	0.0624	0.0547	1	57.23
1	47	0.0190	0.0081	0.0183	0	52.53
2	47	0.0101	0.0074	0.0113	0	56.98

All bounds cover every stored finite state, time, and unilateral action, including deviations at the reported missing pure stages. The full-oracle guard is unused in these principal runs. Profit losses are absolute units; market normalizations are different economies.

and defects are reported in the supplement. A minimum-defect selection at those nodes is not called an exact equilibrium. Nevertheless, the complete dynamic best-response bound remains valid for the resulting profile: its proof does not assume that every fitted one-step game has a pure solution. Moreover, for fixed Markov rivals, randomized or history-dependent own deviations cannot improve on the finite Markov best-response value, because each conditional action mixture is bounded by its largest pure-action payoff. The calculation therefore checks such unilateral deviations as well; it does not compute a mixed-strategy equilibrium profile or a continuous-action game.

R8 adds actor-free neural continuation fitting and classical backward selection for both market normalizations. The neural comparator allocates 480 critic Adam updates per time level rather than R7's 320 critic plus 160 actor updates, with the same L-BFGS cap. Counts and timings reveal the difference between nominal update matching and matched wall time. Both methods receive their own independent full dynamic best responses. The earlier six classical game cases, asymmetric costs, alternative tie rules, boundary frequencies, coarse counterexample, and finer-grid results remain in the supplement and raw record. The finite-game result is a numerically evaluated approximate equilibrium with a specified profit-unit bound. It is not inferred from stationarity of a summed actor loss or the static 1/64 diagnos-

TABLE VII

R8 ACTOR-FREE DYNAMIC-GAME COMPARISONS

Method	Market	Seed	Firm 1	Firm 2	Train (s)	Verify (s)
Classical	1	11	0.0000	0.0000	0.22	6.40
Classical	2	11	0.0000	0.0000	0.23	6.29
Direct	1	11	0.0223	0.0273	37.43	6.09
Direct	1	29	0.0199	0.0282	37.68	6.18
Direct	1	47	0.0223	0.0409	36.03	6.26
Direct	2	11	0.0091	0.0694	40.43	5.75
Direct	2	29	0.0306	0.0185	43.07	6.34
Direct	2	47	0.0206	0.0358	40.47	6.08

The direct method fits two continuation networks with the same class and 480 critic Adam updates, versus R7's 320 critic and 160 actor updates. Classical selection uses no neural approximation. Every method receives a new full dynamic best response. R7 and R8 timings were measured in their recorded environments; cross-environment ratios are not a hardware-controlled speed experiment.

tic. State interpolation, continuous investment, diffusion timing, and continuously monitored exits remain separate terms in a continuous-game error account.

11. SCALABILITY AND THE DIFFUSION TRACE

The original multi-stock consumption-savings specification has additive utility, independent transitions, and no common resource constraint. It decomposes as $V(k) = \sum_j v_j(k_j)$. A separability-aware comparator solves the one-dimensional components, and identical components can be reused. An unreduced sparse grid is not an appropriate sole comparator for that economy.

To test a coupled generator, the new laboratory instead uses

$$dX = (AX + a)dt + \Sigma dW, \quad \sup_a \mathbb{E} \int_0^\infty e^{-\rho t} [-X^\top QX - a^\top a] dt. \quad (49)$$

The drift is a nonnormal dense matrix $A = -.7I + .06Z/\sqrt{d}$, where Z is generated by the fixed model seed $991 + d$. The payoff matrix is $Q = \text{diag}(1, \dots, 2) + .311^\top/d$. The code verifies that $AQ \neq QA$, so the common independent-coordinate reduction of the original benchmark is unavailable. All methods still exploit the declared

quadratic structure. A Riccati solver is therefore the appropriate strong reference, rather than a deliberately unreduced general grid.

For $V = -x^\top Px - c_0$ and $a = -Kx$, policy evaluation solves

$$(A - K - \rho I/2)^\top P + P(A - K - \rho I/2) = -(Q + K^\top K), \quad c_0 = \text{tr}(\Sigma \Sigma^\top P) / \rho. \quad (50)$$

Hamiltonian improvement gives $K = P$. A positive semidefinite P is representable by a signed sum of squared linear neural units; the actor is linear. The exact block laboratory uses a Lyapunov solve for this critic and exact actor improvement. It is an exactly representable NBO subfamily and a matrix-identity test, not evidence that generic stochastic neural optimization has the same convergence rate.

Across dimensions 2, 5, 10, 20, 50 and three stabilizing actor initializations, the block iteration takes six steps. The maximum error in P , in operator norm, is below 10^{-13} relative to an independent algebraic Riccati solver. The policy-evaluation coefficient residual is below 10^{-12} . On the unit Euclidean ball, the actor gap is bounded by $\|K - P\|_2^2$, and the residual bound follows from the operator norm of the quadratic coefficient error. The program saves matrices, iteration traces, and CPU wall times for both methods. These very short, single-process timings are diagnostics and do not establish a fixed-accuracy runtime ranking.

11.1. *Trained, coupled nonquadratic capital economies*

To separate neural scaling from exact quadratic algebra, consider d positive capital stocks and write $Y_i = \log X_i$. Consumption is $c_i = m_i X_i$, with $m_i \in [.02, 2]$, and

$$dY_i = \left[.10 + .15 \tanh((B_d Y)_i) - m_i - \frac{1}{2}(.15^2 + .10^2) \right] dt + .15 dW_i + .10 dW_0, \quad (51)$$

$$r_0(Y, m) = d^{-1} \sum_i (\log m_i + Y_i) - .10 \bar{m}^2, \quad \bar{m} = d^{-1} \sum_i m_i.$$

The dense row-normalized matrix B_d is generated once from a recorded seed. Horizon is one, discount is .04, and terminal utility is $\bar{Y} - .03 d^{-1} \sum_i (Y_i - \bar{Y})^2$. The actual state domain is $\mathbb{R}^d$; no artificial lateral stopping condition is imposed. Dense non-

linear productivity, shared adjustment cost, common noise, and terminal dispersion couple the coordinates. Neither the solver nor its comparator calls a Riccati or Lyapunov routine, and no exact solution is built into the network.

Strict concavity gives a global Hamiltonian maximizer at each queried state. For $p_i = \partial_i v$, its components satisfy

$$m_i^* = \text{clip} \left((dp_i + .2\bar{m}^*)^{-1}, .02, 2 \right),$$

with the upper bound selected when the denominator is nonpositive. The scalar consistency equation for $\bar{m}^*$ is monotone and is solved by bisection. This supplies the same feasible maximization routine to the direct-HJB comparator and to action-gap measurement; it does not supply the value function.

Both networks have two hidden tanh layers of width thirty-two. The critic enforces the terminal payoff exactly. The main design uses three seeds at $d = 2, 5, 10, 20$, 600 Adam updates, and batches of 128 independently sampled state–time points. Exact automatic differentiation supplies the trace. The held-out set contains 512 fixed independent points in $[0, 1] \times [-.5, .5]^d$. A common diagnostic target is residual root mean square at most .025 and maximum queried action gap at most .01. The recorded first crossing is not necessarily sustained; final failures are also reported. The experiment measures real neural approximation and optimization costs rather than exact quadratic identities. It also shows why a diagnostic target is not a welfare certificate. Simulated policy paths can leave the fitted state box; Euler policy-payoff estimates from the initial state need not agree with a critic that has a small sampled residual inside that box. A follow-up expands the fitting and verification box to $[-1.5, .5]^d$ and uses 1,000 updates at dimensions ten and twenty. Both designs, paired policy-payoff samples, and forty/eighty-step simulation outputs are retained. Sampling standard errors do not remove time-discretization bias or certify an optimum. Process high-water memory is labeled as such, not interpreted as an isolated per-network allocation. No high-dimensional uniform residual bound is claimed for these sampled quantities.

TABLE VIII

ACTUAL TRAINING IN DENSE NONQUADRATIC CAPITAL MODELS

Method	d	Worst RMS	Worst gap	Failures	Median seconds
NBO	2	0.0161	2.42×10^{-4}	0/3	2.6554
Direct HJB	2	0.0161	0	0/3	3.4870
NBO	5	0.0131	0.0020	0/3	4.1518
Direct HJB	5	0.0129	0	0/3	5.0391
NBO	10	0.0188	0.0052	0/3	6.8652
Direct HJB	10	0.0168	0	0/3	7.6988
NBO	20	0.0320	0.0114	2/3	12.8180
Direct HJB	20	0.0170	0	0/3	13.5001

Maxima are across three seeds, each evaluated at 512 independent points in $[0, 1] \times [-.5, .5]^d$. The failure criterion is RMS residual above .025 or queried action gap above .01 after 600 updates. Both methods use the same critic architecture and global action routine. Times include training diagnostics, not the subsequent Monte Carlo evaluations. These are sampled diagnostics, not uniform economic bounds.

11.2. Method-paired payoff evidence and arithmetic replay

The R7 follow-up evaluates the coupled-capital policies at dimensions ten and twenty, for all three fixed seeds, both fitting domains, and 40, 80, and 160 Euler steps. The same 512 Brownian paths are used by NBO and the direct comparator within every cell. For path j , define $D_j = J_j^{\text{NBO}} - J_j^{\text{direct}}$. We report $\bar{D}$ and the pointwise Student interval $\bar{D} \pm t_{511, .975} s_D / \sqrt{512}$. Pairing removes the irrelevant component of common path variation from the comparison. The complete set of 36 comparisons is reproduced from the pinned raw arrays in the supplement. The old twenty-dimensional fitting box is exited by essentially all recorded paths. Wider-domain fitting eliminates observed exits in the recorded audits and changes both payoff agreement and method differences. Zero observed exits in 512 paths is not a zero exit-probability theorem. Likewise, these intervals quantify pointwise Monte Carlo uncertainty under a specified Euler scheme. They are neither simultaneous confidence statements across all 36 comparisons nor bounds on time bias, optimal value, or the global differential residual. The experiment identifies a consequential

TABLE IX
 PAIRED PAYOFF DIFFERENCES AT THE FINEST RECORDED EULER LEVEL

Dim.	Seed	Domain	Difference	95% lower	95% upper
10	11	Narrow	-0.03338	-0.03544	-0.03133
10	11	Wide	0.00052	0.00049	0.00054
10	29	Narrow	-0.03094	-0.03250	-0.02938
10	29	Wide	-0.00003	-0.00004	-0.00002
10	47	Narrow	-0.02755	-0.02941	-0.02569
10	47	Wide	0.00000	-0.00001	0.00001
20	11	Narrow	-0.06302	-0.06572	-0.06033
20	11	Wide	0.00084	0.00080	0.00088
20	29	Narrow	-0.01449	-0.01490	-0.01408
20	29	Wide	0.00409	0.00394	0.00424
20	47	Narrow	-0.20161	-0.20211	-0.20111
20	47	Wide	0.00058	0.00053	0.00062

NBO minus direct payoff on the same 512 paths. Full 40/80/160-step comparisons are in the supplement. Confidence intervals quantify Monte Carlo variation only.

coverage failure and its measured correction without redefining the unbounded economic state space as a stopped box.

The one-state consumption certificate has a different evidentiary status. R7 independently replays its previously generated complete leaf cover with `mpmath.iv` at 40 decimal digits, without importing the original interval arithmetic implementation. For the stored seed-11 networks, the NBO policy-loss bound is about .04961 and the direct-HJB bound about .04967. Both meet the stated threshold. This is an *independent interval-arithmetic replay on the previously generated complete cover*, not independent network training, independent cover generation, or a formal machine proof. Together with the new complete three-control cover on the two-state preference grid, it demonstrates two different ways of operationalizing verification. Neither removes the dimensional cost of covering an unrestricted high-dimensional continuous state space.

11.3. Exact and stochastic trace computation

Let Σ include any covariance factor and set $B = \Sigma^\top D^2 v \Sigma$. For independent probes with identity covariance,

$$q = \text{tr}(B), \quad \hat{q} = \frac{1}{K} \sum_{k=1}^K z_k^\top B z_k. \quad (52)$$

Gaussian probes have variance $2\|B\|_F^2/K$. If $r = d - q/2$, then

$$\mathbb{E}(d - \hat{q}/2)^2 = r^2 + \frac{1}{4} \text{Var}(\hat{q}). \quad (53)$$

The extra term depends on the critic. Unbiased trace estimation therefore does not give an unbiased squared-residual objective.

Two conditionally independent probe banks repair this particular bias:

$$\hat{L} = (d - \hat{q}^{(1)}/2)(d - \hat{q}^{(2)}/2), \quad \mathbb{E}[\hat{L} \mid t, x, \xi] = r^2. \quad (54)$$

With dominated differentiation, its gradient is also unbiased. The sample product need not be nonnegative and can have substantial variance, so this is an objective identity rather than a guarantee of stable optimization. Exact low-dimensional derivatives remain the independent check. If the actor or deterministic term depends on the same probes, conditional independence must be re-established; the simple identity cannot be reused automatically.

The relevant computational cost depends on the network evaluation and differentiation cost, Brownian rank, covariance factorization, sample count, optimization iterations, and requested accuracy. Forming a dense Hessian involves quadratic storage; Hessian-vector products can avoid it. Neither operation count by itself is a total-complexity theorem. Active control bounds do not universally create gradient discontinuities, and a noisy trace cannot make a correctly projected box action leave its box. Feasibility and value-derivative accuracy are separate diagnostics.

The independent-bank estimator is also exercised during actual training in the coupled ten-state economy. One implementation uses two independent banks of two Gaussian Hessian-vector probes; the matched single-bank ablation squares a four-probe estimate. Exact traces are used only for the common held-out diagnostics.

All three seeds, objective signs, optimization histories, and policy-payoff samples are saved. The two training procedures perform similarly at the retained covariance scale; unbiasedness alone does not establish a finite-budget advantage. A separate Monte Carlo test checks both the product objective and its parameter gradient against exact trace expressions with declared tolerances. These calculations validate the estimator's execution without identifying a random training objective with a deterministic certificate.

12. VALIDATED EVALUATION AND ACTION IMPROVEMENT

The neural Bellman map must account for two distinct approximation decisions: which continuation values are passed to the next stage, and how tightly the admissible action supremum is bounded. This section supplies an explicit output rule for the first decision and a signed enclosure for the second. Neither construction presumes that a stationary point of the training loss is economically optimal. Both apply to the original preference economy and leave its controls, state law, and stopping convention unchanged.

12.1. *An evaluation rule with a prescribed nodal error*

Let T_h^a denote a positive-weight, one-step policy operator on a finite state grid, and suppose its continuation weights have sum at most $q_h \leq 1$. At a given date, let w be the continuation vector, a the selected feasible policy, $\hat{v}$ the fitted neural critic, and $\tilde{T}_h^a w$ the computed selected-policy target. We distinguish the computation of this target from maximization over actions. In particular, it uses one action at each node, not a table of optimizing labels. For a prescribed tolerance $\tau_h > 0$, define

$$C_h(\hat{v}, a, w)_i = \begin{cases} (\tilde{T}_h^a w)_i, & |\hat{v}_i - (\tilde{T}_h^a w)_i| > \tau_h, \\ \hat{v}_i, & \text{otherwise.} \end{cases} \quad (55)$$

The comparison in the implementation uses an outward upper bound for the difference. A selected entry is assigned the target directly; it is not reconstructed by adding two cancelling floating-point numbers. The indexed corrections, their masks, and the uncorrected neural outputs are saved. The deployed continuation

is therefore a neural representation with a recorded nodal correction, not a raw network.

PROPOSITION 7—Evaluation guard: Suppose $\|\tilde{T}_h^a w - T_h^a w\|_\infty \leq \eta_h$. Then

$$\|C_h(\hat{v}, a, w) - T_h^a w\|_\infty \leq \tau_h + \eta_h.$$

For a frozen sequence of policies over N stages, with exact terminal data, the guarded backward values and the exact values of those policies differ by at most $\sum_{j=0}^{N-1} q_h^j(\tau_h + \eta_h)$. If $Nh = T$, $\tau_h = 0.1h^2$, and $\eta_h = o(h)$, this evaluation perturbation vanishes as $h \downarrow 0$. This statement is a property of the output rule, not a rate for the neural optimizer or an assertion of policy optimality.

The guard gives an implementable meaning to the evaluation component of a vanishing-perturbation condition. It does not supply the action-improvement error, the consistency error of a stopped diffusion approximation, or convergence of an uncorrected network away from the nodes. Its cost can be substantial: in the numerical economy it corrects a large fraction of fitted nodal values. We consequently count the correction memory and compare against an actor-free method with the identical guard. The construction is not evidence of neural compression when almost every node is retained.

12.2. Signed enclosures for continuous actions

Fix a state-time node and a continuation vector w . Write $Q_w(a)$ for its one-step payoff. The action set is the full compact box of the preference economy. Let $B = [\ell, r]$ be an action rectangle with centre c . A natural interval calculation supplies an upper bound $U^{\text{nat}}(B)$, while independent feasible evaluations supply a lower incumbent L .

The continuation is bilinear within state cells and continuous across their internal faces. Reflection changes its one-sided slopes. A stopped transition is different: the implementation switches between interpolated continuation and the specified stopping payoff, which need not agree away from boundary grid points. We call B regular when every cubature transition stays entirely inside, or entirely outside, each

wealth stopping face. Only on such a box do we use derivative information. All intersected interpolation cells and both sides of reflection folds enter the derivative hull.

Suppose $G_j(B) = [g_j^-, g_j^+]$ encloses every one-sided partial derivative with respect to action component j on a regular box. The mean-value upper enclosure is

$$U^{\text{mv}}(B) = \overline{Q}_w(c) + \sum_j \max\{g_j^-(\ell_j - c_j), g_j^-(r_j - c_j), g_j^+(\ell_j - c_j), g_j^+(r_j - c_j)\}, \quad (56)$$

where $\overline{Q}_w(c)$ is an outward point upper bound. The algorithm uses the smaller of this bound and $U^{\text{nat}}(B)$. If $g_j^- > 0$, it restricts the supremum to $a_j = r_j$; if $g_j^+ < 0$, it restricts it to $a_j = \ell_j$. These are verified reductions to a maximizing face, not decisions based on a sampled derivative. For boxes crossing a stopping switch, the natural union enclosure is retained without this reduction.

THEOREM 8—Signed action cover: *Assume that the natural enclosures contain their respective payoffs, that the derivative hulls contain all one-sided derivatives on each regular box, and that feasible point values and arithmetic bounds are outward. Start from a cover of the action box and apply subdivision, reduction to a verified maximizing face, and removal of a box only when its upper bound is at most $L + \varepsilon$. At any interruption, retain the upper bound of every unresolved box. Then*

$$\sup_{a \in \mathcal{A}} Q_w(a) \leq \max\{L + \varepsilon, \max_{B \text{ unresolved}} U(B)\}. \quad (57)$$

The bound remains valid when a depth, work, or time budget is exhausted. If r action coordinates have verified strict derivative signs throughout a regular parent box, maximization over that box reduces to a face of dimension at most $d_a - r$.

The last statement describes a source of computational improvement, not a dimension-free worst-case bound. Under a K -Lipschitz gradient, a regular stationary box of diameter s has mean-value excess of order Ks^2 when its point evaluation error is negligible. Boxes that cross stopping switches, have ambiguous derivative signs, or contain many interpolation cells can remain expensive. Our experiment measures their actual cost and keeps unresolved bounds; it does not assume that every box has favorable structure.

The operational NBO output is now explicit: the frozen policy representation, its guarded continuation vector when used, an independently propagated policy interval, and a complete-action optimal upper envelope. A nodal lookup policy is the object certified in the preference experiment. An off-grid interpolation of that policy is a different deployment rule and is evaluated separately. No online search is charged to a stored lookup table; constructing that table includes every actor, critic, and local-search query.

13. COST, REFINEMENT, AND ECONOMIC ACCURACY

The preceding applications define the original economic scope of NBO. We now evaluate the guarded map and signed cover on the same preference economy, and connect the high-dimensional experiment to an absolute economic benchmark. The comparison retains every earlier failure. In particular, a new upper envelope does not turn an actor-plus-search policy into an actor-only policy, and retraining on a finer grid does not repair the off-grid performance of a previously frozen policy.

13.1. *A tighter certificate for identical frozen policies*

The reference specification has 17×25 state nodes, twenty dates, and the continuous action box $[0.02, 0.30] \times [0, 1.5] \times [-0.45, 0.45]$. The ratios constrain consumption, leverage, and preference adjustment in the declared economy. They are economic primitives, not bounds on an unconstrained optimum. Active consumption or portfolio limits remain meaningful features of this constrained specification; changing them changes the economy.

We freeze the nine previously computed actor-plus-search, direct-plus-search, and search-only policies, including their original seeds. The complete-action upper envelope is recomputed independently of these policies' lower payoff envelopes. The new cover uses a one-step tolerance of 0.0002 and retains every unresolved upper bound. Its output can be shared across policies only because their action set, transition rule, domain, and payoff normalization coincide. Table X reports the resulting all-date, all-node loss bounds. The improvement relative to the historical table concerns the tightness of the verification calculation, not additional training or a change in preferences. The raw actors remain separately reported in the machine-

TABLE X

COMPLETE-ACTION BOUNDS FOR THE SAME FROZEN POLICIES

Policy	Seed	Uniform bound	Centre bound	≤ 0.1
Actor + search	11	0.0687	0.0068	Yes
Actor + search	29	0.0553	0.0050	Yes
Actor + search	47	0.0552	0.0075	Yes
Direct + search	11	0.0465	0.0071	Yes
Direct + search	29	0.0536	0.0052	Yes
Direct + search	47	0.0466	0.0075	Yes
Search only	11	0.0633	0.0113	Yes
Search only	29	0.0655	0.0053	Yes
Search only	47	0.0631	0.0076	Yes

Full 17×25 state grid, twenty dates, unchanged three-control box. The common signed cover took 372.53 seconds; 20 of twenty one-step covers closed at the requested tolerance. Other levels retain all unresolved upper bounds. Bounds are conditional on the stated outward-arithmetic execution model; continuous-state/time transfer is not included.

readable common accounts. Search remains consequential, and these results do not establish an actor advantage over the actor-free comparator. They do show that the earlier failure of a 0.1 upper-bound target need not imply that the frozen hybrid policies themselves are more than 0.1 from the nodal optimum.

For a controlled cost comparison, Table XI gives one-step corner-only and signed covers at three dates, with identical continuation arrays, initial feasible actions, tolerance, depth limit, and work budget. Wall time includes derivative hull construction for the signed cover. The point of this experiment is not to compare two historical machines. Historical training costs are retained in the supplementary total-cost accounts with that qualification. Shared verification is amortized under explicitly stated populations of one, nine, and one hundred policies, and the welfare conversion has a separately counted cost. No claim of neural speed advantage is inferred from this accounting.

TABLE XI

MATCHED ONE-STEP ACTION-COVER COMPUTATION

Cover	Date	Seconds	Q boxes, m.	Bracket	Budget hit
Corner	19	18.67	0.414	0.01336	Yes
Signed	19	9.50	0.066	0.00020	No
Corner	10	19.89	0.440	0.02230	Yes
Signed	10	14.60	0.112	0.00020	No
Corner	0	18.32	0.402	0.03430	Yes
Signed	0	25.84	0.446	0.00173	Yes

Continuation, incumbent, tolerance, and budget are identical within each date. Wall time includes gradient work; a query count is not a hardware-independent speed measure.

13.2. *The whole-domain refinement failure and its repair*

A frozen policy has to be distinguished from a policy refitted on a new grid. The complete historical refinement arrays reveal large losses away from the centre. Their maximum positive reference advantages, locations, quantiles, boundary-region summaries, and action saturation rates are now included in the supplement. The largest historical discrepancy is approximately 2.1234 at high risk aversion and wealth close to the lower stopping boundary. Centre values alone do not summarize that experiment. These discrepancies are feasible-reference lower bounds on policy regret, not upper bounds for the continuous economy.

The new refinement comparison uses genuinely nested spatial grids, 17×25 and 33×49 , with twenty and forty dates, respectively. Each declared actor and actor-free search case is trained afresh. The network width, activation, actor budget, critic budget, and seed set are unchanged across the relevant ablation. We compare the unguarded fitted critic with the indexed guard in (55). For the forty-date case the target tolerance is 0.0000625. Both the fraction of corrected nodes and their storage cost are reported, rather than being attributed to an uncorrected neural approximation. A new complete-action envelope is computed on the fine grid, not interpolated from the coarse certificate. It is applied to all twelve fine-grid policies, including the six unguarded runs. This separates three facts: the failure of a frozen coarse-grid policy; the accuracy achieved by retraining within a declared nodal

TABLE XII

FRESH POLICIES ON THE NESTED FINE GRID

Method	Uniform bound range	Ref. advantage	Mean train s.	Passes
Actor	0.0654–0.0963	0.0932	88.32	3/3
Actor + guard	0.0140–0.0356	0.0155	87.75	3/3
Search	0.0655–0.0963	0.0932	51.07	3/3
Search + guard	0.0349–0.0357	0.0156	47.40	3/3

Full 33×49 state grid, forty dates; seeds 11, 29, and 47. Reference advantage is a lower regret diagnostic against the feasible 175-action reference, not an upper bound. Uniform bounds use a separately computed complete-action fine-grid envelope. The guard is a saved indexed correction, not an uncorrected network.

economy; and the additional error required to transfer a result to a continuously monitored diffusion. The first remains in the record, the second is measured, and the third is not assigned a zero value. Two levels alone do not establish an asymptotic convergence rate.

The deployment specification is a stored state-time action table with its saved neural proposal and continuation correction. Evaluating this table at its nodes requires no online maximizing search. Interpolating the table at off-grid states invokes a different policy rule; the historical experiment shows why this distinction matters. A raw actor without a table or local improvement has a different memory requirement and is not represented by the hybrid loss bounds.

13.3. *A model-specific welfare account*

An absolute utility gap is not a consumption-equivalent percentage when the utility exponent is itself a controlled state. We therefore compute a defined compensating transfer. Hold the saved policy and its state law fixed, leave preference-adjustment costs and the stopping bequest unchanged, and add externally financed consumption so that flow consumption becomes $(1 + e)c_t X_t$. This intervention changes felicity but does not charge its financing to wealth. It is a compensation experiment, not another admissible self-financed policy in the original box.

Let J_e^π be the compensated payoff and U the common optimal upper envelope. A verified lower envelope $J_e^\pi \geq U$ at every active starting node and date implies that this transfer compensates the policy's loss everywhere in the declared nodal economy. We search for a common e using monotonicity of CRRA felicity in consumption and verify the final inequality by outward propagation. Table XIII gives the resulting upper percentages. These conservative transfers can be economically substantial even when the absolute utility bound is below 0.1; the threshold is not described as a universally small welfare loss.

TABLE XIII

SUFFICIENT FINANCED FLOW-CONSUMPTION SUPPLEMENTS

Proposal	Seed	Supplement, percent	Verification seconds
Actor	11	20.93	27.42
Actor	29	19.17	26.86
Actor	47	18.87	26.94
Direct	11	15.33	27.06
Direct	29	18.06	28.34
Direct	47	15.33	27.50
Search	11	20.44	31.32
Search	29	22.46	31.35
Search	47	19.42	31.37

Uniform sufficient compensation over every active starting node and date. State dynamics, the original policy, terminal bequest, and adjustment cost stay fixed. Consumption is externally financed. These are not feasible self-financed policy improvements or common homothetic consumption equivalents.

13.4. A global benchmark for the coupled capital economy

A comparison of two frozen policies does not indicate how far either is from the economic optimum. We supplement the paired simulations by a bound on the original, continuous-time, dense nonlinear capital economy. It does not rely on a Riccati equation, separability of the optimal policy, a state mesh, or a sampled residual.

In the notation of the capital experiment, let Y_i be log capital and $m_i \in [\underline{m}, \bar{m}]$ the consumption ratio. The flow payoff is $\overline{\log m} + \bar{Y} - (\zeta/2)\bar{m}^2$, and the terminal payoff is $\bar{Y} - \chi(\bar{Y} - \bar{Y})^2$, with $\chi = 0.03$. The drift of Y_i is $a + \kappa \tanh((BY)_i) - m_i - (\sigma_I^2 + \sigma_C^2)/2$. Common noise affects the mean but cancels from cross-sectional dispersion. Define

$$b_{\pm} = a \pm \kappa - \frac{1}{2}(\sigma_I^2 + \sigma_C^2), \quad w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)},$$

and let

$$m_0(t) = \text{clip}_{[\underline{m}, \bar{m}]} \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}.$$

This common, deterministic schedule is feasible in the unchanged economy.

PROPOSITION 9—Continuous-economy performance benchmark: *Starting from $Y_i(0) = 0$ for every i , let V^* be the optimal expected payoff of the original d -dimensional capital economy. Define*

$$U = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_+ - m_0(t)] \} dt,$$

$$D_d = \{ \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \}^2,$$

$$L_d = \int_0^T e^{-\rho t} \{ \log m_0(t) - \frac{\zeta}{2} m_0(t)^2 + w(t)[b_- - m_0(t)] \} dt - \chi e^{-\rho T} D_d.$$

Then $L_d \leq J(m_0) \leq V^* \leq U$ for every admissible dense coupling matrix B in the stated model. The inequalities apply to the continuous economy on $\mathbb{R}^d$, not merely to an Euler simulation.

The bound exploits bounded nonlinear drift, concavity of consumption utility, and an explicit feasible policy. Its numerical integration uses outward rectangle sums, interval exponential and logarithm evaluations, and square roots checked by interval squaring. Table XIV reports the computed brackets. Their width is economically material and is not presented as a small regret bound for a learned

policy. They nevertheless provide an absolute benchmark absent from a paired-policy comparison. Simulations of m_0 use the same Brownian paths as the saved neural policies. The resulting differences and confidence intervals concern their common Euler discretization; an unknown simulation bias is not subtracted from the analytical bound to create a neural-policy certificate.

TABLE XIV

GLOBAL BOUNDS IN THE ORIGINAL CONTINUOUS CAPITAL ECONOMY

Dimension	Time panels	Feasible lower	Optimal upper	Width
10	4096	-1.489064	-1.051803	0.437261
10	16384	-1.488883	-1.051998	0.436884
20	4096	-1.489130	-1.051803	0.437327
20	16384	-1.488949	-1.051998	0.436950

Initial log capital is zero in every coordinate; horizon one. The bounds cover every admissible control on $\mathbb{R}^d$. They are not small loss bounds for neural policies. The same feasible schedule is evaluated on the saved Brownian paths in the supplement.

14. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z} \mathbf{1}$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - \nu^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (58)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{ \overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2 \} dt + e^{-\rho T} \{ \bar{Y}_T - \chi \|\Pi Y_T\|_d^2 \} \right], \quad (59)$$

where $v^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1, \rho = .04, a = .10, \kappa = .15, \sigma_I = .15, \sigma_C = .10, \zeta = .2, \chi = .03$, and $[\underline{m}, M] = [.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (60)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. Let $\beta \geq \kappa \|B\|_2$, let s_0 bound the initial cross-sectional standard deviation, and set

$$Q_d = s_0 + \kappa T + \sigma_I \sqrt{(1 - 1/d)T}, \quad (61)$$

$$K(s) = \beta \int_s^T e^{-\rho t} w(t) e^{\beta(t-s)} dt + 2\chi e^{-\rho T} Q_d e^{\beta(T-s)}, \quad (62)$$

$$\mu(t) = \frac{2\{\log[m_0(t)/M] + [M - m_0(t)]/m_0(t)\}}{[M - m_0(t)]^2}. \quad (63)$$

The quantity $\mu(t)$ is the smallest tangent-remainder curvature of log utility on the entire primitive action interval, relative to $m_0(t)$. It is not a Hessian bound estimated from training points. In particular, it can be substantially sharper than the uniform lower curvature M^{-2} .

THEOREM 10—Curvature-controlled policy guarantee: *Suppose $\rho > 0, \zeta, \chi, \kappa \geq 0$, the schedule in (60) is interior, and $\|\Pi y\|_d \leq s_0$. Let $V^*(y) = \sup_m J_y(m)$, where the supremum is over the full original admissible class. Then*

$$0 \leq V^*(y) - J_y(m_0) \leq G_d := \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds. \quad (64)$$

For $\epsilon \geq 0$ satisfying $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$, every admissible policy m^ϕ with $|m_i^\phi(t, Y) - m_0(t)| \leq \epsilon$ for all coordinates, times, and states satisfies

$$0 \leq V^*(y) - J_y(m^\phi) \leq G_d + D_{d,\epsilon}, \quad (65)$$

where, with $A_\beta(T) = \int_0^T e^{\beta t} dt$,

$$D_{d,\epsilon} = \int_0^T \left[\epsilon K(s) + \frac{\epsilon^2 e^{-\rho s}}{2} \{[m_0(s) - \epsilon]^{-2} + \zeta\} \right] ds + \chi e^{-\rho T} \epsilon^2 A_\beta(T)^2. \quad (66)$$

These bounds hold for every initial mean, every initial state with the stated dispersion bound, and arbitrary adapted competitors in the original action box. They require no neural training convergence or discretization error estimate.

The proof uses synchronous state comparison, an exact discounted mean-state identity, and completion of a quadratic bound on an arbitrary policy deviation. It is provided in the appendix. The result is a model-specific numerical principle: curvature limits the benefit from changing consumption, whereas a spectral bound limits the propagation of that change through dense nonlinear production. Their balance yields a scalar integral certificate for a high-dimensional policy. Bounded nonlinearity, the additive diffusion structure, and the chosen felicity are essential assumptions, rather than generic properties of all HJB equations.

14.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (67)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded

proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (68)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

14.2. Computable scope and cost

An SVD is used only to propose a scalar λ . Positive pivots in an outward interval $LDL^\top$ calculation for $\lambda I - B^\top B$ certify the spectral majorant. A failed pivot triggers enlargement, not acceptance of the SVD output. The remaining integrals use outward rectangle sums, polynomial exponential and logarithm bounds, and square-root endpoints checked by interval squaring. For a fixed majorant with verified pivots, this certificate requires $O(d^3)$ arithmetic operations and $O(d^2)$ storage for the matrix check, followed by $O(N)$ scalar quadrature work. This is not a dimension-independent training theorem or a worst-case bound on the number of attempted majorants.

The guarantee holds already at initialization. Training is intended to improve the feasible policy inside the tube, not to manufacture its certificate. The anchor itself has the tighter bound G_d and requires no training. A comparison omitting it would overstate the role of the actor. Neither a small $G_d + D_{d,\epsilon}$ nor faster actor evaluation proves a net economic advantage over that benchmark.

15. TRAINING, DEPLOYMENT, AND CERTIFIED CAPITAL ACCURACY

We execute the preceding construction on the unchanged capital economy in dimensions ten, twenty, and fifty. Each dimension has three fixed initialization

seeds, 11, 29, and 47. The actor and critic have two hyperbolic-tangent hidden layers of width 32. The actor radius is $\epsilon = .1$. The critic incorporates the terminal payoff exactly and includes a generic neural correction to the discounted mean and dispersion terms. This is an approximation architecture, not an exact invariant solution class.

15.1. *A fixed protocol with strong comparators*

Each fit uses 600 critic updates, batch size 128, and the same state draws and initialization seeds across methods. The critic learning rate is .0015; the actor learning rate is .002. Two independent Gaussian trace banks, with two probes each, are actually used in the product residual objective. Negative sample losses are retained. Training states have $t \in [0, 1]$ and each log capital in $[-1.5, .5]$. Exact-Hessian diagnostics at 128 held-out points do not enter the continuous certificate.

The direct-tube comparator removes the actor and globally maximizes the same critic Hamiltonian on the identical time-dependent tube. The direct-full comparator instead maximizes on the full primitive box $[.02, 2]^d$. Both exploit the scalar monotone mean-consumption equation and use 44 bisection steps. They receive the same critic architecture, sample stream, trace banks, and update count. Their additional maximization work is counted. This is a matched-update design, not a claim that unequal wall times are equal compute budgets. The untrained feasible schedule m_0 is a fourth comparator.

There are 27 fitted policies, including all seeds and both direct alternatives. Every saved actor is deployed without local action search. The full deployment cost includes the inward scalar schedule guard; direct deployment also includes critic differentiation and bisection. Separate process runs record peak resident memory, training time, query counts, and optimizer histories. The raw proposals, guarded deployment routines, and the scalar benchmark are not treated as interchangeable policies.

15.2. Continuous-time accuracy and its economic scale

Table XV reports outward bounds for the anchor and for every policy in the declared tube. The initial cross-sectional standard deviation is at most the stated s_0 , while the initial mean is unrestricted. No state-space covering or unknown diffusion-transfer term enters these numbers. The complete-action comparison class remains the original primitive box, not the actor tube.

TABLE XV

CONTINUOUS-ECONOMY BOUNDS FOR EVERY DEPLOYED TUBE POLICY

d	s_0	Anchor bound	Tube bound	Flow supplement (%)
10	0.00	0.01875	0.05743	6.04
10	0.25	0.02152	0.06184	6.52
10	0.50	0.02455	0.06649	7.02
20	0.00	0.02553	0.06725	7.11
20	0.25	0.02881	0.07219	7.65
20	0.50	0.03235	0.07740	8.22
50	0.00	0.02754	0.07009	7.42
50	0.25	0.03097	0.07518	7.98
50	0.50	0.03465	0.08054	8.57

$\epsilon = .1$; 16,384 outward quadrature panels. Initial mean is unrestricted. The optimum allows all adapted controls in $[\cdot, .02, 2]^d$. Supplement is externally financed and uses an outward logarithmic-utility conversion. Printed upper endpoints are rounded upward. Bounds are conditional on the documented arithmetic contract.

The certificate is recomputed at 4,096 and 16,384 quadrature panels. The finer enclosure, verified matrix majorants, and all initial-dispersion and radius variants are saved. This refinement concerns the numerical integration of a proved continuous-economy bound, not a grid approximation to the state process. The spectral verification and scalar integration costs are charged explicitly. The certificate is common to the actor and direct-tube policy class; it is not a seed-specific estimate of achieved regret.

For an interpretable utility scale, consider an externally financed proportional supplement to every consumption flow, holding the policy-induced state law, ad-

justment cost, and terminal payoff fixed. A supplement e raises discounted payoff by

$$\log(1 + e) \frac{1 - e^{-\rho T}}{\rho}.$$

Thus an outward upper value of $\exp\{\rho(G_d + D_{d,\epsilon})/(1 - e^{-\rho T})\} - 1$ is a sufficient flow-consumption compensation. This is a defined externally financed welfare comparison, not a claim that the same transfer is self-financing or that all economic models share a common utility percentage.

15.3. *What is learned, and what is guaranteed*

Table XVI reports the actual cost of training and deploying the policies. The supplement reports every seed, rather than only an average of successful cases. The structural certificate already holds for the initial zero-output actor; therefore the time to its first certificate is not its training time. The anchor has a tighter guarantee and no neural training cost. These facts are central to the interpretation, not omitted ablations.

We separately compare frozen policies on common Brownian paths with 40, 80, and 160 Euler steps and 512 paths per dimension. All fine increments, terminal states, and pathwise payoffs are stored. Table XVII gives paired actor-minus-comparator differences at the finest level. The reported intervals cover Monte Carlo variation only; they exclude time-discretization bias and are not used to sharpen Theorem 10. Multiple seeds and resolutions are retained without a multiplicity-adjusted discovery claim.

This experiment supplies a trained multilayer NBO policy, a strong direct comparator, a deployed-policy specification, and an independent continuous-economy guarantee in one coupled nonlinear model. It does not establish that the actor is universally preferable to a structured feasible schedule, that constant-step optimization converges, or that the same certificate applies to endogenous preference formation. The latter economy retains the separate nodal and continuous-transfer accounts developed above. Likewise, the dynamic-game deviation bounds retain their stated finite-game scope.

TABLE XVI

MEASURED COST OF THE TRAINED AND DEPLOYED POLICIES

d	Method	Training (s)	Training + bound (s)	Deployment (ms)
10	Schedule	0.000	0.113	0.053
10	NBO actor	5.284	5.397	1.703
10	Direct tube	6.426	6.539	5.724
10	Direct full	6.366	–	4.140
20	Schedule	0.000	0.115	0.058
20	NBO actor	5.600	5.715	1.766
20	Direct tube	6.753	6.868	6.143
20	Direct full	6.658	–	4.550
50	Schedule	0.000	0.129	0.054
50	NBO actor	6.259	6.387	1.840
50	Direct tube	7.837	7.966	7.711
50	Direct full	7.929	–	6.073

Training is the mean of three separate seed runs; deployment evaluates 256 states and includes the inward schedule guard where applicable. Matrix verification plus scalar quadrature is charged in full to each certified method; no shared-cost amortization is used. The schedule needs no training and has the tighter guarantee. Direct-full training time is recorded, but no tube certificate is assigned to it. Detailed per-run memory and failures appear in the supplement.

16. CONCLUSION

Neural Bellman Operators organize computation around policy evaluation, feasible improvement, and independent economic verification. Recursive utility, endogenous preferences, sophisticated temporal selves, and dynamic strategic interaction remain part of that economic formulation. The representation is neural, but an economic claim concerns the policy actually deployed and the error that has been bounded.

The guarded map specifies how a nodal evaluation tolerance is enforced and records the indexed corrections it requires. Signed derivative enclosures reduce complete-action verification work without dropping unresolved boxes or differentiating across a stopping discontinuity. Their numerical contribution is identified using identical frozen policies and matched one-step comparisons. Fresh training

TABLE XVII

 PAIRED ACTOR-MINUS-COMPARATOR PAYOFF DIFFERENCES

d	Comparator	Mean difference	Lower 95%	Upper 95%
10	Schedule	0.001299	0.001280	0.001319
10	Direct tube	0.000861	0.000821	0.000902
10	Direct full	0.000965	0.000916	0.001014
20	Schedule	-0.001954	-0.002043	-0.001865
20	Direct tube	0.007563	0.007408	0.007719
20	Direct full	0.071241	0.067724	0.074758
50	Schedule	-0.000647	-0.000707	-0.000587
50	Direct tube	0.012037	0.011900	0.012173
50	Direct full	0.412308	0.402341	0.422276

512 common paths, 160 Euler steps. Each row averages the pathwise payoff differences of the three fixed seed fits before computing a Monte Carlo interval. These are not intervals over a population of training seeds and do not include Euler bias. No simulation result is substituted into the continuous-time proof.

and recertification on genuinely nested grids are kept distinct from the full-domain failure of interpolated coarse policies. The compensation account translates utility accuracy into a stated financed-transfer experiment rather than a misleading common percentage.

The dense capital experiment now has a feasible-policy lower benchmark and a continuous-time optimal upper benchmark on the original state space. Those historical absolute brackets are distinct from the new continuous-time neural-policy bound in Theorem 10. The latter uses the original capital economy and full adapted comparison class, with a schedule-centred actor architecture. Paired simulations remain discretized policy comparisons. Likewise, the complete-action preference certificates do not yet bound all continuous-state, continuous-time, and stopping-monitoring defects. The evidence does not establish that actors outperform the strongest low-dimensional search and projection alternatives; those comparisons remain in the record. The contribution is a specified neural Bellman computation with verified evaluation and improvement outputs, measured costs, and separate economic domains for each conclusion.

APPENDIX A: PROOFS

A.1. *Proof of Theorem 2*

Extend the aggregator Lipschitz-continuously outside its invariant interval for this comparison argument. The extension leaves all policy utilities and the candidate residual unchanged. For a nonnegative number c , define $d_c(t) = Be^{L(T-t)} + cA_L(T-t)$. Then $d_c(T) = B$, $d_c(t) \geq B$, and $-d'_c = Ld_c + c$. Since a spatially constant shift leaves the generator unchanged and f is L -Lipschitz in utility,

$$F[v + d_c] \geq F[v] - d'_c - Ld_c = F[v] + c, \quad F[v - d_c] \leq F[v] + d'_c + Ld_c = F[v] - c.$$

Because $F[v] = r^a[v] - g^a[v] \in [-e - \eta, e]$, $v + d_{e+\eta}$ is a supersolution and $v - d_{e+\eta}$ is a subsolution. They dominate the required stopping data on every boundary face. A spatially constant shift also preserves an exactly satisfied homogeneous reflected boundary condition. Comparison proves (15).

Apply the same argument to the fixed-policy operator $F^a[v] = -v_t - \mathcal{H}^a[v]$, with $c = e$, to obtain (14). The triangle inequality gives (16), and policy admissibility gives its nonnegative lower bound. The construction uses $L \geq 0$ and $d_c(t) \geq B$ specifically to cover lateral stopping boundaries; it does not incorrectly discount their errors as if every path stopped only at T . For nonsmooth candidates the same argument applies when their residual inequalities are verified in the viscosity sense. The theorem's differential implementation, however, requires the stated smooth candidate. Q.E.D.

A.2. *Exact policy improvement*

Suppose $v = V^a$ solves the fixed-policy problem and a feasible $\tilde{a}$ satisfies $\mathcal{H}^{\tilde{a}}[v] \geq \mathcal{H}^a[v]$ pointwise. Then $F^{\tilde{a}}[v] \leq 0$ with the same boundary payoff. Policy comparison gives $V^{\tilde{a}} \geq V^a$. This is a policy-space statement. A single gradient step on a domain-averaged actor loss does not automatically meet its pointwise premise. The measured action gap and policy-evaluation errors are retained precisely because approximate neural improvement can violate that premise.

A.3. Proof of Theorem 3

For a fixed state and action, let z and $\tilde{z}$ solve (17) for w and $\tilde{w}$. Strict monotonicity of $z - hf(z)$ gives

$$|z - \tilde{z}| \leq \frac{|P_h^a(w - \tilde{w})|}{1 + \lambda h} \leq q_h \|w - \tilde{w}\|_\infty.$$

The same monotonicity implies order preservation. Taking the maximum over actions preserves both order and the Lipschitz constant. Banach's fixed-point argument supplies the two fixed points on the invariant complete value domain.

Now $\|v - v_h^a\| \leq \|v - T_h^a v\| + q_h \|v - v_h^a\|$, which gives the first bound. Also $\|v - T_h v\| \leq \delta + \epsilon$, giving the second. The triangle inequality gives the third. All norms are over every state of the specified finite model. An omitted action changes T_h and must be accounted for as a separate error, not hidden in δ . Q.E.D.

A.4. Proof of Theorem 4

First, (18) gives

$$\|v_h - v_h^*\|_\infty \leq \frac{\delta_h + \epsilon_h}{1 - q_h} = \frac{1 + \lambda h}{\lambda} \frac{\delta_h + \epsilon_h}{h} \longrightarrow 0.$$

It remains to identify the limit of the exact monotone scheme. Stability makes the upper and lower half-relaxed limits finite. At a strict local maximum of the upper limit minus a smooth test function, select nearby maximizing grid points. Monotonicity compares the neighboring scheme values to that test function plus the vanishing contact offset. Local consistency then gives the viscosity subsolution inequality. The corresponding minima give the supersolution inequality for the lower limit. Consistent boundary data and the assumed barriers give the required boundary inequalities. Comparison orders the upper limit below the lower limit. The reverse order follows by definition, so the two limits coincide with the unique continuous viscosity solution. The vanishing distance between v_h and v_h^* transfers that convergence to the neural arrays. The discrete policy-value conclusion follows from the third bound. Interpreting a discrete policy as a continuous-time implementation additionally requires convergence of its controlled interpolation and payoff; the theorem does not infer this merely from convergence of its state arrays. This is the

usual monotone-scheme selection argument, with the neural approximation error made explicit; see [Barles and Souganidis \(1991\)](#), [Crandall et al. \(1992\)](#). *Q.E.D.*

A.5. Finite-horizon Bellman error accumulation

Let T_n and T_n^a be the time- n Bellman and fixed-policy maps, with Lipschitz constants q_n (which may exceed one). For arrays v_n , define $\delta_n = \|v_n - T_n^a v_{n+1}\|$ and $\epsilon_n = \|T_n v_{n+1} - T_n^a v_{n+1}\|$. If the terminal error is B_N , backward induction gives

$$\|v_n - v_n^*\| \leq \left(\prod_{j=n}^{N-1} q_j \right) B_N + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) (\delta_k + \epsilon_k).$$

The policy-evaluation bound replaces $\delta_k + \epsilon_k$ by δ_k . Their sum bounds regret. Empty products equal one. For the implicit Lipschitz recursive step, $q_n \leq (1 - hL)^{-1}$; for additive exponential discount, $q_n = e^{-\rho h}$. Thus a per-step error $o(h)$ and consistent terminal data suffice on a fixed finite horizon, without asserting a stationary contraction in an inapplicable utility regime.

A.6. Derivation of the temporal-self condition

A self at (t, x) replaces a^* by a fixed feasible local action a for duration ϵ and then returns to a^* . For each exponential component, the deviation value is

$$\mathbb{E} \left[\int_t^{t+\epsilon} e^{-\rho_j(s-t)} u(a) ds + e^{-\rho_j \epsilon} W_j(t + \epsilon, S_{t+\epsilon}^a) \right].$$

Itô expansion gives its first-order increment relative to $W_j(t, x)$ as $\epsilon[u(a) + (W_j)_t + \mathcal{L}^a W_j - \rho_j W_j] + o(\epsilon)$. Weighting the two components and subtracting their evaluation equations under a^* leaves

$$\epsilon\{u(a) - u(a^*) + (\mathcal{L}^a - \mathcal{L}^{a^*})V\} + o(\epsilon).$$

Nonpositive first-order gain for every feasible action is exactly (45). This derivation assumes the smoothness and local integrability needed for the expansion; it is an equilibrium deviation condition, not an assertion of optimal commitment over a finite interval. The homothetic derivation follows by integrating $e^{-[\rho_j - b(r-m)]s}$ and applying the consumption first-order condition. *Q.E.D.*

APPENDIX B: PROOFS FOR THE GUARDED AND SIGNED MAP

B.1. *Proof of Proposition 7*

At a replaced node the output equals the computed target. At any other node its distance from that target is at most τ_h . The triangle inequality gives the one-step conclusion. For a frozen policy sequence, let e_n be the sup-norm difference between the guarded and exact policy values at date n . Positivity of the transition weights and their mass bound imply $e_n \leq \tau_h + \eta_h + q_h e_{n+1}$, with $e_N = 0$ for exact terminal data. Backward induction gives the geometric sum. As $N = T/h$ and $q_h \leq 1$, the bound is at most $T(0.1h + \eta_h/h)$ under the stated schedule. No property of the optimizer was used. The assertion concerns the guarded nodal values. An off-node neural interpolant, an improved policy, and a diffusion approximation each require their separate error account. In fixed-precision arithmetic, $\eta_h = o(h)$ is a maintained refinement condition, not a fact that can be inferred as h tends to zero indefinitely. $\square$

B.2. *Proof of Theorem 8*

On a regular box, the payoff restricted to a line segment is continuous and piecewise continuously differentiable, with finitely many interpolation and reflection knots, or is the limit of such segments. Integration of its almost-everywhere derivative yields

$$Q_w(a) - Q_w(c) = \int_0^1 \nabla Q_w(c + s(a - c)) \cdot (a - c) ds.$$

Every partial derivative is included in its interval hull. Maximizing the resulting interval products proves (56). Taking the minimum with another valid upper enclosure preserves validity. If a partial derivative is strictly positive throughout the box, integration along that coordinate shows that replacing it by its upper endpoint cannot lower the payoff; the negative case is analogous. Repeating the argument proves face domination and the dimension statement.

A subdivision replaces a parent by children whose union contains it. A face reduction replaces a set by a subset with the same supremum. Removal occurs only after that supremum has a bound no larger than the incumbent plus tolerance. Feasible point evaluations can only increase the incumbent, so an earlier removed

box stays bounded by the final incumbent plus tolerance. The remaining boxes keep their upper bounds without any success inference from an interrupted search. Induction over all operations proves (57). At a stopping switch the line-segment continuity premise can fail; the implementation therefore uses the natural union enclosure and does not invoke the derivative argument there.

For the local quadratic observation, a K -Lipschitz gradient with a zero-gradient point in the box has norm at most Ks throughout a box of diameter s . Integration over a segment of length at most s bounds the variation by Ks^2 . This conditional observation does not extend to an arbitrary switching box, nor does it assert that all nodes possess such a box. $\square$

B.3. Derivative hulls for the preference economy

Write $y = \log X$, and let F be the continuation interpolant evaluated at a cubature transition. On an interior leg, its action derivatives are

$$\partial_c F = -hF_y, \quad \partial_p F = F_y \{h(0.06 - 0.04p) + s_z\}, \quad \partial_\theta F = hF_u r_u,$$

where s_z is the stored portfolio shock coefficient and $r_u \in \{-1, 1\}$ is the reflection slope; both slopes enter the hull at a fold. The flow derivatives are

$$h \exp\{(1-u)y - u \log c\}, \quad 0, \quad -hk\theta.$$

Outside a wealth stopping face, clipped wealth is constant and F is the exact stopping payoff. Its y derivative with respect to the transition is zero, while

$$F_u = \frac{\exp\{(1-u)y\} \{1 - (1-u)y\}}{(1-u)^2}.$$

Every crossed bilinear cell contributes its one-sided derivative range, restricted by the other coordinate's interval. Interval evaluation includes these ranges and the flow terms. A leg whose wealth interval meets both sides of a stopping switch disables the signed bound for that box. Point and interval evaluations share the stated nodal primitives but are computed independently of the actor's derivative graph.

B.4. Proof of Proposition 9

Let $\bar{m} = d^{-1} \sum_i m_i$. Jensen's inequality gives $d^{-1} \sum_i \log m_i \leq \log \bar{m}$. Moreover, $d^{-1} \sum_i \tanh((BY)_i) \leq 1$. The mean-state drift is consequently no larger than $b_+ - \bar{m}$. The dispersion penalty at the terminal date is nonnegative and can be dropped for an upper bound. Fubini's theorem applied to the integrated expected mean state and terminal mean state yields the kernel $w(t)$. The Brownian terms have zero mean; bounded controls and bounded nonlinear drift ensure the required integrability. Thus every admissible policy has expected payoff at most

$$\int_0^T e^{-\rho t} \max_{m \in [\underline{m}, \bar{m}]} \{ \log m - \zeta m^2 / 2 + w(t)(b_+ - m) \} dt.$$

The integrand is strictly concave in m . Its derivative is $1/m - \zeta m - w(t)$, whose positive root gives $m_0(t)$ before clipping. This proves the upper bound.

For the feasible common schedule, the expected mean-state drift is at least $b_- - m_0(t)$. Define the normalized dispersion norm $\|Z\|_{2,d}^2 = \mathbb{E}[d^{-1} \sum_i Z_i^2]$. Centred common controls and common noise vanish from the dispersion equation. For numbers in $[-1, 1]$, their cross-sectional variance is at most one. Hence Minkowski's inequality bounds the norm of the integrated centred nonlinear drift by κT . The centred idiosyncratic Brownian part has norm $\sigma_I \sqrt{(1 - 1/d)T}$. A second application of Minkowski and squaring give the upper bound D_d on expected terminal dispersion. Subtracting $\chi e^{-\rho T} D_d$ from the lower mean-payoff calculation proves $J(m_0) \geq L_d$. Feasibility gives $J(m_0) \leq V^*$. The derivation never restricts the dense matrix B beyond the stated bounded-tanh form. $\square$

B.5. Compensation and arithmetic scope

For each realized state and positive consumption, CRRA felicity is increasing in consumption, including when $u > 1$ and its level is negative. The financed supplement leaves the state law and every other payoff term unchanged. Its expected payoff is therefore nondecreasing in e . Positivity of the nodal transition operator propagates a lower interval payoff at each date. If its lower endpoint exceeds the optimal upper envelope at every active starting node and date, the corresponding financed supplement compensates the loss throughout that nodal economy. Bisec-

tion searches for a sufficient supplement; only the final verified inequality provides the stated upper bound.

All implemented interval conclusions use the inherited binary64 execution contract, with outward elementary operations and separately bounded transcendental evaluations. They concern exact real interpretations of stored primitive coefficients and grid nodes. They are not a formal verification of the interpreter, hardware, or all library internals. In the capital bound, square-root seeds are accepted only after interval squaring verifies both endpoint inequalities, and the time integral is bounded by interval rectangles rather than by an adaptive quadrature tolerance. Random stress tests are regression evidence, not substitutes for these inclusion arguments.

APPENDIX C: PROOF OF THE CONTINUOUS-CAPITAL GUARANTEE

All norms in this proof use the normalization $d^{-1} \sum_i z_i^2$. This normalization is important: the spectral norm of B remains the ordinary Euclidean operator norm, while the stochastic moment of cross-sectional dispersion is of order one rather than order d .

C.1. Logarithmic tangent remainder

For $a > 0$ and $x > -a$, twice integrating the second derivative gives

$$\frac{x}{a} - \log(1 + x/a) = x^2 \int_0^1 \frac{1-r}{(a+rx)^2} dr.$$

For $x \geq 0$ the integral decreases as x increases. For $x < 0$ it is at least $1/(2a^2)$, its value at zero. Consequently, for every $m \in [\underline{m}, M]$ and interior $a < M$,

$$\log(m/a) - \frac{m-a}{a} \leq -\frac{\mu(a)}{2}(m-a)^2, \quad \mu(a) = \frac{2\{\log(a/M) + (M-a)/a\}}{(M-a)^2}. \quad (69)$$

On the smaller interval $|m-a| \leq \epsilon < a$, the same integral also yields

$$0 \leq \frac{m-a}{a} - \log(m/a) \leq \frac{(m-a)^2}{2(a-\epsilon)^2}. \quad (70)$$

These are global inequalities on the specified intervals, including their endpoints;
no empirical concavity check is used.

C.2. Synchronous comparison and baseline moments

Construct Y^m and Y^0 on the same probability space with identical initial state and Brownian drivers, where the second process uses m_0 . With $\Delta Y = Y^m - Y^0$ and $\Delta m = m - m_0 \mathbf{1}$, additive noise cancels. Since componentwise hyperbolic tangent is one-Lipschitz,

$$\|\Delta Y_t\|_d \leq \int_0^t \{\beta \|\Delta Y_s\|_d + \|\Delta m_s\|_d\} ds.$$

Gronwall's inequality followed by Minkowski's integral inequality implies

$$(\mathbb{E} \|\Delta Y_t\|_d^2)^{1/2} \leq \int_0^t e^{\beta(t-s)} r(s) ds, \quad r(s) = (\mathbb{E} \|\Delta m_s\|_d^2)^{1/2}. \quad (71)$$

All terms are integrable because controls and the nonlinear production term are bounded. For the baseline, common consumption, common drift constants, and common noise vanish under Π . Furthermore, $\|\Pi \tanh(BY)\|_d \leq 1$ and $\mathbb{E} \|\Pi W_T\|_d^2 = (1 - 1/d)T$. Minkowski's inequality therefore gives

$$(\mathbb{E} \|\Pi Y_T^0\|_d^2)^{1/2} \leq \|\Pi y\|_d + \kappa T + \sigma_I \sqrt{(1 - 1/d)T} \leq Q_d. \quad (72)$$

This estimate is uniform in the initial mean. It does not suppose that the optimal policy is symmetric or that the dense economy separates.

C.3. An exact payoff-difference identity

Put $h(Y) = \overline{\tanh(BY)}$. Integrating the mean-state equation inside the discounted flow and terminal mean, and interchanging integrable expectations and time integrals, yields

$$\begin{aligned} J_y(m) - J_y(m_0) &= \int_0^T e^{-\rho t} \mathbb{E} \left[\overline{\log(m_t/m_0(t))} - w(t) \overline{\Delta m_t} - \frac{\zeta}{2} \{\bar{m}_t^2 - m_0(t)^2\} \right] dt \\ &\quad + \kappa \int_0^T e^{-\rho t} w(t) \mathbb{E} [h(Y_t^m) - h(Y_t^0)] dt \\ &\quad - \chi e^{-\rho T} \mathbb{E} \left[2 \langle \Pi Y_T^0, \Pi \Delta Y_T \rangle_d + \|\Pi \Delta Y_T\|_d^2 \right]. \end{aligned} \quad (73)$$

The martingale mean terms have zero expectation. The schedule satisfies $m_0(t)^{-1} - w(t) - \zeta m_0(t) = 0$. Thus all terms linear in $\overline{\Delta m}_t$ cancel. Equation (69) bounds the first integral above by $-\frac{1}{2} \int e^{-\rho t} \mu(t) r(t)^2 dt$; the remaining mean-adjustment square is nonpositive and may be dropped in an upper bound.

Next, $|h(Y^m) - h(Y^0)| \leq \|B\|_2 \|\Delta Y\|_d$. Equations (71) and (72), Cauchy–Schwarz, and a change in integration order bound the potentially positive production and terminal-dispersion terms by $\int_0^T K(s) r(s) ds$. Dropping the nonpositive terminal square, we obtain

$$J_y(m) - J_y(m_0) \leq \int_0^T \left[K(s) r(s) - \frac{1}{2} e^{-\rho s} \mu(s) r(s)^2 \right] ds \leq \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds.$$

Taking the supremum over the original adapted controls proves (64). Attainment of the supremum is not needed.

For a tube policy, $r(s) \leq \epsilon$. Reverse the payoff difference in (73). Equation (70) and $\mathbb{E}(\overline{\Delta m})^2 \leq \mathbb{E}\|\Delta m\|_d^2$ bound the consumption and adjustment loss by

$$\frac{\epsilon^2}{2} \int_0^T e^{-\rho s} \{[m_0(s) - \epsilon]^{-2} + \zeta\} ds.$$

The production and terminal cross terms contribute at most $\epsilon \int K(s) ds$. The terminal square now has positive sign and is bounded by $\chi e^{-\rho T} \epsilon^2 A_\beta(T)^2$ using (71). This proves $J_y(m_0) - J_y(m^\phi) \leq D_{d,\epsilon}$. Adding (64) proves (65). Feasibility supplies the nonnegative lower bound. Q.E.D.

C.4. Interval spectral and deployment checks

For the stored exact-real matrix B , outward products and sums enclose each entry of $B^\top B$. The Schur complement recurrence for an $LDL^\top$ factorization is enclosed entry by entry. If every pivot interval has strictly positive lower endpoint, every exact pivot is positive, so $\lambda I - B^\top B$ is positive definite. Hence $\|B\|_2 < \sqrt{\lambda}$. The proposed numerical singular value is not part of this proof; an unsuccessful interval factorization supplies no spectral bound.

If $\ell \leq m_0(t) \leq u$, then $\text{up}(u - \epsilon) \geq m_0(t) - \epsilon$ and $\text{down}(\ell + \epsilon) \leq m_0(t) + \epsilon$. Every finite output after projection onto (68) therefore meets the theorem’s tube condition

under the declared arithmetic contract, regardless of errors in the neural proposal. The proof does not verify the interpreter or processor. The arithmetic assumptions, overflow checks, saved weights, and input-clock convention remain explicit components of the computational claim.

APPENDIX D: REPLICATION AND SCOPE OF NUMERICAL EVIDENCE

Integrated R8 candidate. The present article incorporates the following immutable repository inputs.

R7 source: 5bab4649728860fb655c5371f3705787f10b4a78

R7 evidence: b1409008625f550195ce9ead468391ad2449a240

R7 advisory report: c63f6817f498e1cfe9a653139f7a25bf44a4418a

The preceding main manuscript and supplement are preserved by exact Git blob identity in the R8 archive. R8 source, all executed development configurations, raw arrays, independent certificates, method-paired comparisons, generated tables, and compilation logs belong to the candidate itself. The response answers every B1–B7 and M1–M9 comment of the latest report. The later unexecuted R7 development protocol is not imported as evidence. A table manifest checks each printed number against its input file; the execution manifest identifies the exact R8 source and candidate workflow. Earlier review and revision branches remain unchanged.

The revision directory contains the executable block core, the experiment program, the regression tests, raw numerical outputs, and a response-to-referee document. The reviewed manuscript, supplement, and historical comments are preserved byte-for-byte in the archive, and the review branch remains unchanged. The new main manuscript is authoritative for the revised equations. The archive is not compiled into the revised article and its unsupported numerical statements are not reused as evidence.

The inherited R2 execution contains 29 recorded experiments and eight regression tests, all retained and rerun. The R6 record adds nine primary stopped-consumption runs, six architecture/update sensitivities, six full-domain certificates, six principal NDU neural runs, sixteen NDU reference sensitivities, twenty-four dense-capital primary fits, six stochastic-trace fits, twelve coverage follow-ups, and six principal

dynamic-game solves. Pilot failures and coarse-game counterchecks are retained separately rather than counted as successful primary runs. The original referee diagnostics continue to concern their pinned original sources.

The initial development summary and the corresponding script are preserved separately. Inspection found a sign error in the *reported* quadratic residual formula even though the Lyapunov solutions agreed with Riccati values. The formula was corrected, its acceptance test strengthened to include that residual, and the complete experiment suite rerun. The wealth stopping boundary was also imposed explicitly at all NDU time levels before the final run. The earlier results are marked superseded diagnostics rather than silently relabelled as final evidence.

The R6 directory contains complete source, generated tables, model and numerical arrays, optimizer histories, certificate covers, tests, and a provenance manifest. The replication workflow materializes the revision source, independently reruns its computations, and commits the resulting evidence only to the designated R6 candidate branch; an exact-source snapshot and raw-result hashes identify each execution. Local execution, remote replication, and independent referee acceptance remain different facts. No historical GPU benchmark, unexecuted deep-BSDE comparison, unrestricted neural Cournot equilibrium, or empirical identification result is inferred from the computations reported here.

4.0.0.1. R9 revision record. The authoritative R9 source descends from the review of the R8 candidate, dated 3 October 2026. Its new material, all raw runs, complete-action arrays, full-domain refinement summaries, proofs, response, and execution identities are stored under `revisions/2026-10-03-r9`. The archived R8 main text and supplement preserve the pre-revision scholarly record. Every inherited section and label is retained. The new response addresses the R8 report’s B1–B7 and M1–M10 comments. Earlier replication and response descriptions in this historical record refer to their respective R6–R8 rounds; historical workflow success is not substituted for a run of the current source.

4.0.0.2. R10 revision record. The current source descends from the complete R9 evidence commit. Section 14 and its proof give the continuous-capital guarantee; Section 15 reports the complete training and deployment comparison. Ex-

act previous roots, all new source and results, the response to the 3 October 2026 R8 advisory report, and the current execution identities are stored under revisions/2026-10-04-r10. Earlier numerical accounts and unsuccessful experiments remain unchanged.

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